

L_RAG: A Controlled Ablation Study of Vietnamese Legal Retrieval-Augmented Generation

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Abstract

Legal question answering must retrieve evidence that is relevant, complete in its statutory context, valid at the relevant time, and traceable to an authoritative source. We evaluate L_RAG, a Vietnamese legal retrieval-augmented generation pipeline that combines E5 dense retrieval, BM25, a knowledge-graph overlay, reciprocal-rank fusion, second-stage reranking, and citation-grounded generation. On a fixed 500-question benchmark (400 answerable and 100 unanswerable questions), we report five controlled ablation analyses grouped into four run families rather than a pooled factorial comparison: embedding text; dense, hybrid, and graph retrieval as one joint retrieval/graph run family; reranking within a shared candidate pool; and Base versus chain-of-thought (CoT) prompting. Metadata improves document-level success at depth 10 by 4.5 percentage points but reduces exact-chunk MRR. The tested BM25+RRF hybrid loses 5.98 Recall@10 points to dense retrieval; its recorded median total rises from 5.483 to 31.538 seconds under an accounting convention that includes amortized BM25 precomputation and is not online request latency. One-hop graph expansion loses 11.81–12.40 Recall@10 points, chiefly by replacing seed evidence. In a common 30-candidate pool, Jina reranking improves Recall@10 by 9.23 points and nDCG@10 by 11.85 points over RRF-only. Under fixed context, CoT decreases GPT-4o-mini Token F1 by 1.57 points. Retrieval is scored against gold chunk identifiers; lexical answer overlap, template-based answerability behavior, structural citation fields, and recorded latency are separate diagnostics. These are benchmark-bounded engineering findings: they do not establish legal correctness, citation entailment, or faithful reasoning.

1 Introduction

Legal information access is an evidence-sensitive task: a useful answer must identify the govern-

ing text, preserve the context in which a provision operates, and allow the reader to inspect its source. These requirements are especially important for Vietnamese law, where relevant rules are distributed across laws, decrees, decisions, and circulars connected by hierarchical, referential, and temporal relationships. This report studies L_RAG, a Vietnamese legal Retrieval-Augmented Generation (RAG) prototype that combines corpus structuring, dense and sparse retrieval, a legal knowledge graph, reranking, and citation-grounded answer generation. The central question is not whether adding more components makes a pipeline appear more sophisticated, but which components measurably improve evidence retrieval and answer behavior under controlled comparisons.

1.1 Background and Motivation

Keyword search remains valuable in legal research when a user already knows an article number, document title, or exact statutory phrase. In ordinary question answering, however, users often describe a situation in everyday Vietnamese rather than reproduce the terminology of an official instrument. Dense retrieval can bridge this lexical gap by matching semantic content, and large language models can turn retrieved passages into readable answers. Neither capability alone is sufficient for dependable legal assistance.

Legal documents are structured and interdependent rather than independent blocks of text. The meaning of a clause may depend on its article heading, parent document, neighboring provisions, referenced instruments, or effective date. A passage that is topically similar can therefore be incomplete, superseded, or disconnected from the rule needed to answer the question. Moreover, a fluent generated response can conceal missing evidence or attach a citation that does not support the adjacent claim. These risks motivate a pipeline in

which retrieval units preserve provenance, graph relations expose legal structure, and generation is constrained by inspectable evidence.

L_RAG provides a common implementation and evaluation setting for comparing these choices. In particular, it enables controlled tests of metadata-enriched embeddings, dense versus dense-sparse retrieval, one-hop graph expansion, second-stage rerankers, and base versus explicit reasoning prompts. This evidence-driven framing is necessary because a component that is intuitively appropriate for legal text may still reduce exact evidence coverage or impose an unacceptable latency cost in a fixed pipeline.

1.2 Problem Statement

Given a natural-language Vietnamese legal question q , a corpus of legal documents $\mathcal{D}$, and retrieval units $\mathcal{U}$ derived from those documents, the system must select a compact evidence set $E \subset \mathcal{U}$ and generate an answer a whose material claims can be traced through chunk, provision, and document identifiers. A legal time or scope constraint can further restrict applicability when the corresponding overlay is explicitly supplied. The task is difficult because four failure modes interact:

1. **Lexical mismatch.** A user’s vocabulary may differ from the formal wording of the applicable provision, which weakens exact keyword matching.
2. **Structural fragmentation.** Chunking can separate a clause from its article header, parent instrument, definitions, exceptions, or adjacent provisions, causing retrieval of locally relevant but incomplete evidence.
3. **Relationship and validity blindness.** Semantic similarity alone does not represent citation, amendment, replacement, authority, or effective-date relationships between instruments.
4. **Ranking and generation error.** Relevant evidence may be present in a candidate pool but ranked outside the final context window; the generator may then omit support, refuse an answer unnecessarily, or produce an unsupported statement.

These failures have practical consequences: the system may cite the wrong instrument, overlook a controlling provision, present outdated material,

or produce an answer that cannot be audited. The research problem is therefore to determine how semantic, lexical, structural, and generative components can be combined while measuring both evidence quality and operational cost. In this report, success is evaluated through benchmark-bounded retrieval, answer overlap, answer/refuse behavior, structural citation, completion, and latency diagnostics; it is not treated as certification of legal correctness.

1.3 Project Objectives

The project pursues the following measurable objectives, each linked to an implemented module and a report artifact:

1. **Build traceable retrieval units from Vietnamese legal data.** Normalize documents into provisions and chunks with stable document, provision, and chunk identifiers and retained metadata. Corpus scale and graph integrity are reported in Sections 3 and 3.9.
2. **Implement comparable evidence-retrieval paths.** Provide dense E5 retrieval, Vietnamese-tokenized BM25 retrieval, Reciprocal Rank Fusion (RRF), and graph-based filtering or expansion behind a shared pipeline. Their behavior is measured using Recall@ k , Hit@ k , MRR@ k , nDCG@ k , precision, and latency in Sections 5 and 6.
3. **Model legal structure and inter-document relations.** Build document, provision, and chunk nodes with containment, reading-order, and cross-document edges, while retaining provenance and verification gates. The schema and traversal role are specified in Sections 4 and 4.4.
4. **Improve final-context ranking under a fixed evidence ceiling.** Compare RRF-only ordering with mMiniLM, Qwen3, BGE, and Jina rerankers over the same 30-candidate pool and measure their ability to retain and promote gold evidence into the top 10, together with reranking latency.
5. **Support grounded generation and explicit abstention.** Construct answers from the selected context, expose source markers, and instruct the generator to refuse when evidence is insufficient. Generated outputs are assessed

with lexical-overlap, template-based answerability, citation-structure, completion, and latency diagnostics, with their limitations stated explicitly.

6. **Create a reproducible evaluation basis.** Use a frozen 500-question benchmark—400 answerable and 100 unanswerable cases—with evidence identifiers and saved run artifacts. Paired comparisons and bootstrap confidence intervals isolate the tested component within each ablation family.

1.4 Scope of the System

The implemented scope covers the offline preparation of Vietnamese legal documents, structural parsing into provisions and chunks, dense and sparse index construction, knowledge-graph construction, candidate retrieval and fusion, graph-assisted context selection, second-stage reranking, prompt construction, citation-grounded generation, and offline evaluation. The experimental scope is limited to the available corpus and frozen benchmark, the recorded index and graph versions, the tested retrieval budgets, and the models and prompts documented in Section 5. The reported graph ablation tests bounded structural expansion with broad validity filtering; it does not test the separate query-time temporal-validity or authority overlay.

The project does not claim exhaustive coverage of Vietnamese law, real-time synchronization with authoritative legal databases, resolution of every amendment or relationship-direction ambiguity, or validation of the generated answers by legal experts. The benchmark is partly synthetic and contains relatively small multi-hop and cross-document slices. Automatic overlap and citation-marker metrics do not establish legal correctness, completeness, or citation entailment. L_RAG is therefore a research prototype and evaluation platform, not a source of legal advice or an autonomous decision-making system. Any deployment affecting rights, obligations, or remedies must verify the cited text and its current legal status and retain qualified human review.

1.5 Main Contributions

This work makes five main contributions:

1. A processed Vietnamese legal corpus with document–provision–chunk provenance, comprising 151,624 retained document

records, 1,386,267 provisions, and 1,513,376 retrieval chunks.

2. A typed legal graph containing structural reading-order links and 883,256 retained cross-document relationship records plus 19,763 external stubs, together with quarantine records, a verified-edge view, temporal and authority overlay interfaces, and bounded context expansion. The overlay interfaces are documented system capabilities, not effects tested by the reported graph ablation.
3. A modular end-to-end pipeline that integrates multilingual dense retrieval, Vietnamese BM25, RRF, four second-stage rerankers, and evidence-constrained answer generation.
4. A 500-question Vietnamese legal QA benchmark with answerability, category, answer type, reference answer, and document-, provision-, and chunk-level evidence provenance.
5. Five controlled analyses grouped into four run families, separating component value from architectural intent. The experiments show that metadata primarily improves document-level identification, the tested hybrid and graph settings trade away exact evidence coverage, reranking provides the clearest retrieval gains, and explicit reasoning prompts do not automatically improve generated-answer overlap.

1.6 Report Organization

The report follows a Problem → Data → Methodology → Evaluation → Discussion progression. Section 2 positions the problem within RAG, legal IR, hybrid retrieval, and graph-augmented retrieval. Section 3 describes the processed corpus and QA benchmark, while Section 4 presents the L_RAG architecture and methodology. Section 5 defines the implementation, metrics, controls, and reproducibility protocol. Section 6 presents the controlled ablations and their results. Section 7 interprets the findings and states the technical and ethical boundaries, and Section 8 summarizes the evidence and identifies future work.

2 Background and Related Work

This section positions L_RAG at the intersection of retrieval-augmented generation, legal informa-

tion retrieval, hybrid ranking, knowledge-graph retrieval, reranking, and citation-grounded generation. It first establishes the standard RAG workflow, then identifies the legal properties that ordinary text retrieval does not capture. The purpose is not to claim that one paradigm solves legal QA by itself, but to make explicit which prior ideas motivate each component and which evaluation check tests the corresponding design assumption. Table 1 provides the design-to-evidence roadmap before the individual research strands are discussed.

2.1 Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) (Lewis et al., 2020) supplements a language model’s parameters with documents retrieved at query time. A standard pipeline contains five logical stages: (1) represent the query and retrieve candidates from an indexed corpus; (2) select, order, and possibly expand a bounded context; (3) serialize that context with provenance into a prompt; (4) generate an answer conditioned on the prompt; and (5) check whether the answer is supported by, and correctly attributes, the supplied evidence. The external context can reduce dependence on stale parametric knowledge and gives a user sources to inspect, but it does not guarantee that the selected passages are sufficient or that the generated claims follow from them.

The basic RAG abstraction is useful for L_RAG because it separates three possible failure locations: the retriever can miss the governing passage, the context builder can omit its parent or exception, and the generator can fail to use or cite the evidence. Advanced methods such as Self-RAG (Asai et al., 2024) and Corrective RAG (Yan et al., 2024) add self-critique or retrieval-quality assessment and correction. They motivate separating retrieval quality from answer behavior, but they do not by themselves encode the formal hierarchy, amendment history, or temporal validity of Vietnamese legal instruments.

2.2 Legal Information Retrieval

Legal Information Retrieval (Legal IR) is a specialized area of information retrieval and legal informatics (Locke and Zuccon, 2022). Legal texts differ from open-domain documents in ways that change both indexing and evaluation:

1. **Hierarchical structure.** An instrument is organized into parts, chapters, articles, clauses,

and items. The meaning of a chunk can depend on its parent heading or a neighboring exception.

2. **Precision requirements.** A missing qualifier, number, or negation can change the interpretation of a rule even when the retrieved paragraph is topically correct.
3. **Authority and validity.** The Constitution, laws, decrees, decisions, and circulars occupy different authority levels, and the applicable instrument depends on effective dates and amendment status.
4. **Explicit dependencies.** A provision may cite, amend, replace, or repeal another instrument. These relationships are evidence about context, not ordinary lexical similarity.

Legal-BERT (Chalkidis et al., 2020) and Lawformer (Xiao et al., 2021) show how domain-adapted transformers can represent legal language. Such encoders improve semantic matching, but a vector alone does not state whether an article is current, whether a reference is authoritative, or which sibling chunk supplies a missing exception. L_RAG therefore retains structured identifiers and graph relations alongside the embedded text.

2.3 Dense, Sparse, and Hybrid Retrieval

Dense bi-encoders are effective for paraphrase and vocabulary variation, whereas sparse methods such as BM25 (Robertson and Zaragoza, 2009) reward exact term overlap and remain strong for document codes, names, numbers, and technical terminology. E5 (Wang et al., 2022) motivates the dense component in L_RAG; BM25 supplies a lexical complement when a question uses a statutory phrase verbatim. The two methods fail differently, but complementarity does not imply that every fusion configuration will improve the final ranking. Adding a modality can change candidate ordering, tokenization behavior, and latency, so the trade-off must be established empirically.

Reciprocal Rank Fusion (RRF) (Cormack et al., 2009) combines ranked lists without requiring calibrated scores. For a candidate d and retrieval systems $m \in M$,

$$\text{RRF}(d) = \sum_{m \in M} \frac{1}{k + r_m(d)}, \quad (1)$$

where $r_m(d)$ is the rank assigned by system m and k is a smoothing constant. L_RAG uses $k = 60$ in

Table 1: Legal RAG requirements, corresponding L_RAG design choices, and evaluation checks. Automatic checks diagnose system behavior; they do not replace expert assessment of legal correctness.

Requirement	L_RAG design response	Evaluation check
Traceable provenance	Stable document, provision, and chunk identifiers; source markers retained through generation	Gold-ID Recall@ k /Hit@ k and citation-marker validity
Context preservation	Metadata-enriched embedding text plus containment and reading-order graph edges	Chunk-only versus chunk+metadata ablation and seed-preserving/lost-evidence analysis for graph expansion
Legal applicability	Authority and effective-date metadata, validity timeline, quarantine, and direction-verification gates	Schema/build audit; the current graph ablation does not estimate the temporal overlay’s effect, and semantic validity remains a human-label gap
Complementary access	E5 dense retrieval, Vietnamese-tokenized BM25, and RRF candidate fusion	Paired DenseOnly versus HybridOnly Recall, MRR, nDCG, and latency
Evidence prioritization	Second-stage reranking of a shared candidate pool before context construction	Candidate Recall@30 ceiling versus final Recall/MRR/nDCG@10 and reranker latency
Grounded failure behavior	Evidence-constrained prompting, source markers, and explicit insufficient-context response	Answerability decisions, citation presence/structure, completion, and latency, reported separately from legal correctness

its RRF configurations. Because fusion can help one query while harming another, the ablation reports paired recall, rank, and latency deltas rather than assuming that hybrid retrieval is an unconditional improvement.

2.4 Knowledge-Graph-Augmented Retrieval

Knowledge graphs represent entities and explicit relationships as triples (e_1, r, e_2) . GraphRAG (Edge et al., 2024) demonstrates graph-derived community context for query-focused synthesis over large collections. L_RAG uses a different, legal-structure graph: it connects a document to its provisions, a provision to its chunks, adjacent chunks in reading order, and one instrument to another through citation, amendment, replacement, expiry, guidance, or suspension edges.

This structure can restore context that a fixed-size vector chunk omits. It also introduces failure modes that do not occur in a simple vector index: incorrect edge direction, stale validity fields, and a traversal policy that replaces high-confidence seeds under a fixed context budget. L_RAG therefore stores direction-verification, validity-timeline, and authority metadata in separate auditable structures. The reported G-LRAG ablation, however, tests only bounded one-hop structural expansion with broad validity filtering; it does not activate the query-time temporal-validity or authority overlay. Its result is evidence about that tested policy, not graph retrieval in general.

2.5 Reranking and Citation-Grounded Generation

Cross-encoder rerankers score the query and candidate passage jointly rather than comparing two independently encoded vectors. Passage reranking with BERT (Nogueira and Cho, 2019) motivates this second-stage design: it can improve the order of evidence at the cost of additional inference time. L_RAG compares pairwise multilingual rerankers from mMiniLM, BGE/FlagEmbedding (Chen et al., 2024), and Jina (Jina AI, 2024), together with the causal-language-model-based Qwen3 Reranker (Zhang et al., 2025), over a shared 30-candidate pool. Holding the pool fixed makes a positive effect interpretable as evidence promotion rather than new evidence access.

Citation-grounded generation adds source identifiers or provision references to the answer format and asks the generator to abstain when context is insufficient. Work on attributed generation distinguishes whether citations are present from whether they entail the associated claims and collectively cover the answer (Gao et al., 2023). These controls improve inspectability, but a citation marker is not the same as claim-to-source entailment. The report consequently treats citation presence, structural coverage, and refusal markers as observable diagnostics and does not convert them into legal-correctness labels.

2.6 Design Requirements for Legal RAG

The literature above implies six requirements for a defensible legal RAG system. Table 1 maps each requirement to an implemented L_RAG design response and to the evaluation evidence used in this report. This mapping prevents an architectural feature from being presented as a benefit before its effect has been measured.

L_RAG represents these requirements through structured corpus records, graph/overlay interfaces, frozen per-question artifacts, and paired analyses. Not every represented capability is activated in every experiment. The remaining evidence gaps are therefore explicit: the current automatic checks do not validate temporal applicability, claim-level citation entailment, or the substantive legal correctness of generated answers. The dataset and benchmark that make the implemented checks reproducible are described next.

3 Legal Dataset and Benchmark Construction

The experiments use two linked data products: a processed Vietnamese legal corpus and a question-answer benchmark with explicit evidence provenance. The ingestion path preserves the hierarchy of an instrument instead of flattening all text into unrelated passages. This section records the transformations, quality gates, and final statistics needed to reproduce the evaluation denominators.

3.1 Legal Corpus and Raw Data Sources

The raw corpus is derived from the Hugging Face dataset [thinhng0/vietnamese-legal-documents](https://huggingface.co/thinhng0/vietnamese-legal-documents) (Ngô, 2026). According to its dataset card, the corpus was crawled from the [Vietnamese Government Legal Document Portal](https://www.vietnam.gov.vn/) using Scrapy. The project uses three upstream configurations: metadata, content, and relationships, containing document metadata, raw HTML bodies, and directed document-to-document relations, respectively.

For reproducibility, all experiments use a frozen local snapshot processed by dataset pipeline version 2.0.0 and reconciled on 2026-07-06. The snapshot includes metadata, HTML content, and relationship records. Relationship labels originate from the source portal metadata and are not generated by an LLM.

The snapshot is not treated as a live legal service.

Table 2: Normalization rules applied before indexing.

Field	Rule and audit behavior
Identifier	Preserve the source ID as a string and use it as the join key across corpus artifacts.
Dates	Parse DD/MM/YYYY, emit ISO YYYY-MM-DD, preserve missing values, and flag invalid dates.
Document type	Map source labels to the controlled document-type vocabulary and authority rank while retaining the raw value.
Status	Normalize validity metadata to canonical status fields and preserve unknown values rather than inferring them.
Facets	Normalize authority, legal field, sector, and scope using versioned vocabularies while retaining raw source values.
HTML text	Merge content rows by document ID, hash the raw HTML, extract normalized text, and retain extraction-quality flags.
Duplicates	Quarantine duplicate relationship triples and keep one deterministic accepted record per edge key.

All corpus-size and reconciliation counts are reported once in Section 3.9; they describe the frozen evaluation package and do not imply complete coverage or current legal validity of all Vietnamese legal instruments.

3.2 Dataset Normalization and Cleaning

Normalization converts heterogeneous source records into deterministic fields before structuring and indexing. Source identifiers are preserved as strings and used consistently across metadata, content, relationship, provision, chunk, and graph artifacts. Text fields are Unicode-normalized to NFC and cleaned using deterministic whitespace rules without free-form rewriting. Portal dates in DD/MM/YYYY format are converted to ISO YYYY-MM-DD; missing values remain null, while malformed or invalid dates are recorded as quality flags.

Document types are mapped to a controlled legal-document vocabulary and to seven authority ranks, while validity metadata is normalized into canonical status groups such as *active*, *expired*, *partial*, *suspended*, *future*, and *unknown*. Issuing authority, legal field, sector, and scope are likewise mapped to versioned vocabularies while preserving their original source values for auditability.

Raw HTML is not embedded directly. Content records sharing the same document identifier are deterministically merged, hashed, and converted to normalized text using `bs4+lxm1+cleanrules@1`. The pipeline retains provenance information and quality flags so that extraction and normalization decisions can be audited. The main normalization rules are summarized in Table 2.

Cleaning is intentionally conservative. Unresolved relationships are retained as auditable references rather than silently discarded. Records with missing required metadata, unknown document type, invalid issue-date conditions, quarantined relationship endpoints, duplicate relationship triples, or self-loops are excluded from the accepted corpus and recorded in the corresponding quarantine ledger.

3.3 Relationship Processing and External References

Relationship processing uses the source-provided relationships records from the portal crawl. Each record identifies a source document, a target document, and a Vietnamese relationship label. The pipeline normalizes document identifiers and maps source labels to a canonical relation vocabulary while preserving the original record for provenance.

These source records define only document-to-document legal relations such as citation, amendment, replacement, suspension, and related-document links. Containment and reading-order edges are constructed deterministically by the text-structuring layer: `DOCUMENT_HAS_PROVISION` links each accepted document to its parsed provisions, `PROVISION_HAS_CHUNK` links each provision to its emitted chunks, and `PROVISION_NEXT/CHUNK_NEXT` connect siblings in source order. The report therefore distinguishes source legal relations from derived structural graph edges.

Graph edges follow the convention that `src_id -[REL]-> dst_id` reads as “source REL target.” Relation groups whose direction can be deterministically resolved from the source semantics, including citation, guidance, supplementation, amendment, and related-document relations, are marked as direction-verified. Relations involving replacement, validity, or suspension remain explicitly unverified when their direction cannot be established reliably from the available metadata. Such edges are retained for audit but are not treated as authoritative directional evidence by downstream graph reasoning.

Targets that resolve to accepted corpus documents are linked directly to their canonical document identifiers. References to documents outside the frozen corpus are represented as `ExternalStub` nodes rather than fabricated retrievable documents. These stubs preserve the

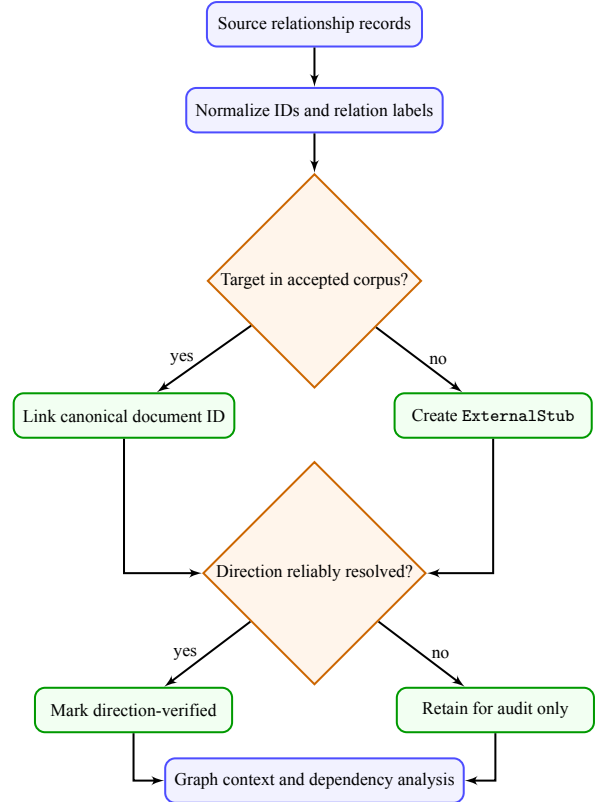


Figure 1: Relationship-processing flow for resolving targets and verifying relation direction.

Table 3: Graph relation sources in the processed corpus.

Relation family	Source and interpretation
<code>DOCUMENT_HAS_PROVISION</code>	Deterministically constructed containment edge from document to parsed provision.
<code>PROVISION_HAS_CHUNK</code>	Deterministically constructed containment edge from provision to indexed chunk.
<code>PROVISION_NEXT, CHUNK_NEXT</code>	Deterministically constructed reading-order edges between siblings.
<code>based_on, cites, amends, replaces, etc.</code>	Source dataset document-to-document legal relations after canonicalization and direction checks.

existence and provenance of external references while remaining excluded from normal text retrieval. Figure 1 summarizes the target-resolution and relation-direction workflow.

The resulting relation inventory supports both structural context recovery and cross-document dependency analysis. Authority and temporal-validity reasoning is applied only when the required metadata and relation direction have passed the corresponding verification checks.

3.4 Quarantine and Data-Quality Verification

Quality verification is applied before records enter the retrieval index or graph. The pipeline

Table 4: Quarantine reason tags in the frozen v2 corpus. Reason counts are non-exclusive tags and may therefore sum to more than the number of quarantined rows.

Stream	Reason tag	Count
Documents	missing_issuer	1,789
Documents	unknown_type	8
Documents	future_issue_date	2
Relationships	duplicate_edge	7,652
Relationships	source_quarantined	4,781
Relationships	target_quarantined	2,030
Relationships	self_loop	277

checks metadata completeness, document type and date validity, structural parentage, text availability, relationship endpoints, duplicate edges, self-loops, and relation-direction status. Failed metadata or relationship records are written to quarantine ledgers with explicit reason tags. Missing or unusable body text is recorded separately in `text_provenance.jsonl` and produces no retrievable provisions or chunks.

Table 4 reports the quarantine reason tags. Row-level accepted and quarantined totals are reported in Section 3.9.

Unresolved external targets are not treated as quarantine failures; they are retained as `ExternalStub` records. Similarly, relations with unverified direction remain auditable but are excluded from authoritative directional reasoning, while unknown validity metadata is preserved rather than inferred.

This conservative policy favors traceability over corpus size: records with unreliable metadata or structure are excluded from retrieval rather than silently repaired. The resulting coverage trade-off is discussed in Section 7.

3.5 Legal Text Structuring

The corpus follows the hierarchy *Document* $\rightarrow$ *Provision* $\rightarrow$ *Chunk*. Documents retain source-level metadata, provisions represent citable legal units such as articles, items, sections, and preambles, and chunks are the units indexed for retrieval. Chunk-level retrieval metrics are therefore evaluated against `chunk_id` identifiers.

Chunking is provision-bounded: text from different provisions is never merged into the same chunk. Provisions whose extracted text does not exceed the hard limit remain as a single chunk. The pipeline uses a maximum of approximately 1,200 estimated tokens, corresponding to `CHUNK_MAX_CHARS = 4,800` under the determin-

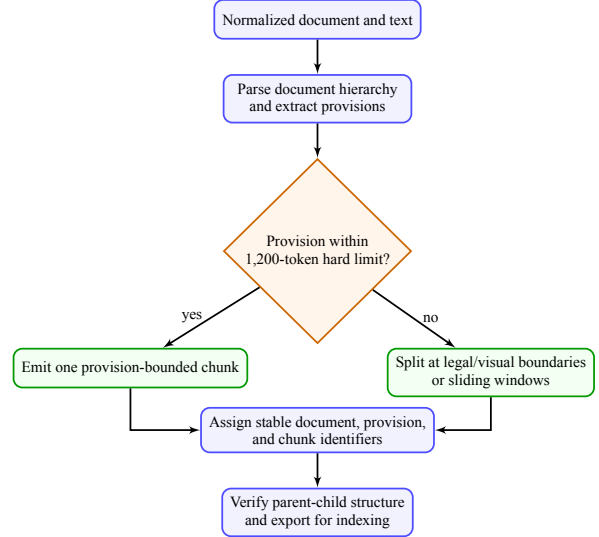


Figure 2: Deterministic document-to-provision-to-chunk structuring workflow.

Table 5: Chunk-length distribution in the frozen corpus. Token counts are deterministic estimates from the text-structuring pipeline.

Measure	Min	P25	Median	P75	P95	Max
Characters	3	333	715	1,495	3,597	4,800
Estimated tokens	1	83	178	373	899	1,200

istic estimate of four characters per token.

Longer provisions are split within their parent provision. The splitter first prefers legal and visual boundaries such as *khoán*, numbered paragraphs, list items, and table-like breaks. When these boundaries are insufficient, it falls back to a paragraph-aware sliding-window strategy targeting approximately 900 estimated tokens per chunk with about 130 tokens of overlap. All emitted chunks remain subject to the 1,200-token hard maximum. Figure 2 summarizes the deterministic structuring and chunk-emission path.

Document titles, provision headings, and citation metadata are retained at the parent levels rather than duplicated into the base chunk record. They are joined with chunk text only when constructing metadata-enriched retrieval representations. Stable document, provision, and chunk identifiers preserve provenance throughout indexing, retrieval, and evaluation.

The final structured corpus contains 1,386,267 provisions and 1,513,376 chunks. Most provisions remain unsplit: 1,340,934 produce a single chunk, while 172,442 chunk records originate from provisions that required splitting. Structural coverage and parent-child consistency are verified automat-

ically before export; this verification concerns corpus structure rather than human assessment of legal completeness.

This representation allows retrieved chunks to be matched directly to benchmark evidence while preserving their originating provision and document. It also enables neighboring context to be recovered without losing source provenance.

3.6 QA Benchmark Construction

The benchmark contains 500 Vietnamese legal questions: 400 answerable questions with reference evidence and 100 questions verified as unanswerable within the frozen corpus. Benchmark generation is implemented in `notebooks/archive/gen-qa.ipynb`. The pipeline samples seed provisions from the frozen v2 corpus, constructs related evidence bundles containing up to five provisions, and uses `gemini-3.1-flash-lite` to generate user-style Vietnamese legal question-answer pairs.

The generation prompt provides the selected evidence bundle, a code-defined generation plan, and structured output constraints. It discourages trivial document-centric questions and instead targets practical legal information needs. A separate generation path produces validity and cross-document cases involving amendment history, partial validity, and interactions between legal instruments.

Unanswerable items are represented as an answer type rather than a separate question category. They are generated by requesting plausible legal details for which no supporting evidence is available in the frozen corpus, such as a specific amount, office, form, procedure, or condition. To verify these cases, the notebook gives the verifier the candidate question, the proposed insufficient-context answer, and the evidence bundle from which the question was generated, then requires a pass only when the evidence is insufficient and the empty gold set is appropriate. The frozen-corpus claim is operational: within the benchmark pipeline, no supporting document, provision, or chunk identifier from the frozen corpus is assigned, and the verifier confirms that the supplied evidence does not answer the question. It is not a human legal proof that no possible support exists outside the frozen snapshot. Their reference answer is normalized to “Không có đủ thông tin trong đoạn luật.”, and their document, provision, and chunk gold identifiers are empty by construction. In the final benchmark, the 100 unanswerable items consist of 93

Table 6: QA generation, verification, diversity filtering, and deduplication counts.

Stage	Count
Sampled documents	515
Sampled provisions with valid chunks	1,697
General candidates	660
Nuance candidates	94
Candidates submitted to verification	754
Passed verification	546
Failed verification	208
Removed by diversity filtering	46
Removed as exact duplicates	0
Final benchmark	500

single-hop, 5 legal-validity, 1 cross-document, and 1 multi-hop questions.

Verification is performed in a separate model call using `gemini-3.1-flash-lite`. Each candidate is checked against its evidence for answer correctness, evidence sufficiency, unsupported claims, category consistency, and difficulty consistency. A candidate is rejected if any required verification criterion fails. The verifier also assigns one of three difficulty labels: *easy*, *medium*, or *hard*.

The verifier pass rate is 72.41% (546/754). Diversity filtering is applied after verification, limiting the benchmark to at most two QA items per provision and fifteen per document. The final records retain a stable `qa_id` across retrieval, generation, and post-hoc evaluation.

3.7 Question Categories and Answer Types

Question categories describe the evidence pattern required to answer a question. *Single-hop* questions rely on one local passage; *citation* questions require identifying the relevant legal authority; *legal-validity* questions depend on temporal status or validity; *multi-hop* questions require multiple linked provisions; and *cross-document* questions require evidence spanning related legal instruments.

Each benchmark item is assigned exactly one primary category. When multiple labels are plausible, the most diagnostic evidence pattern is selected, prioritizing legal-validity, cross-document, multi-hop, citation, and then single-hop.

Answer type is recorded separately as *abstractive*, *boolean*, *extractive*, or *unanswerable*. Difficulty is also independent of category and answer type. The verifier assigns *easy* to local single-fact cases, *medium* to questions requiring multiple facts, linked provisions, cross-document evidence, or validity reasoning, and *hard* to cases involving interacting conflict or partial-validity signals.

Table 7: Benchmark category and answer-type inventory. The benchmark contains 400 answerable and 100 unanswerable questions.

Category	Count	Answer type	Count
Single hop	265	Abstractive	199
Citation	106	Boolean	123
Legal validity	96	Extractive	78
Multi hop	21	Unanswerable	100
Cross-document	12		

The category counts sum to 500, while the answer-type counts separate the 400 answerable questions from the 100 unanswerable questions.

3.8 Ground-Truth Evidence

Each benchmark item stores its question, category, answer type, reference answer, and a structured evidence object containing document, provision, and chunk identifiers:

$$Q_i = (\text{qa_id}, \text{question}, \text{category}, \text{answer_type}, \text{reference_answer}, \text{ground_truth}). \quad (2)$$

For each answerable question, the gold chunk set G_i is the relevance unit used for $\text{Recall}@k$, $\text{Hit}@k$, $\text{MRR}@k$, and $\text{nDCG}@k$. Document and provision identifiers are retained to support document-level and structural analyses in addition to exact chunk ranking. For unanswerable questions, $G_i = \emptyset$ by construction and these cases are excluded from retrieval-metric denominators.

Gold identifiers are derived from code-defined generation plans rather than free-form model output. For answerable questions, the generation plan specifies the supporting provision identifiers. A minimal-evidence tagger then maps these provisions to chunk-level evidence partly by retaining chunks whose token overlap with the reference answer is at least 0.25. If no chunk satisfies this threshold, all chunks belonging to the planned provision are retained to avoid assigning an answerable question an empty gold set. A validation step rejects answerable records that lack document, provision, or chunk identifiers, after which the verifier checks that the reference answer is supported by the supplied evidence.

Across the 400 answerable questions, the mean gold cardinality is 1.163 documents, 1.383 provisions, and 1.375 chunks, corresponding to 550 gold-chunk observations in total. Multi-gold cases arise when a question requires multiple provisions,

Table 8: Gold-set cardinality in the 500-question benchmark.

Gold field	0 IDs	1 ID	2-3 IDs	4+ IDs
Documents	100	341	59	0
Provisions	100	310	79	11
Chunks	100	308	83	9

when a supporting provision is split into several chunks, or when the overlap-based tagger retains all chunks of a planned provision.

The term *gold evidence* is used operationally for the benchmark’s reference identifiers rather than as expert-adjudicated legal ground truth. Because chunk assignment partly depends on reference-answer token overlap, alternative provisions or broader authority sets may also support a valid answer, but the automatic retrieval metrics credit only the identifiers included in the benchmark.

3.9 Dataset Statistics and Validation

Before export, the pipeline reconciles source, normalized, structured, and indexed counts. Every document points to zero or more provisions; every indexed chunk points to one provision and one document; every resolved legal edge points to existing endpoints; and every external reference points to a stub rather than an invented target. Missing-text, orphan, quarantine, and unknown-validity counts are retained in the audit ledger even when the final retrieval index excludes them.

The validation identities are:

$$\begin{aligned} N_{\text{raw}} &= N_{\text{accepted}} + N_{\text{quarantined}}, \\ N_{\text{accepted_relationships}} &= N_{\text{resolved}} + N_{\text{external_target}}. \end{aligned} \quad (3)$$

These counts make the corpus inventory auditable. They also explain why a graph or strict validity filter can reduce available evidence: the system prefers a visible missing or uncertain record to an unverified relation presented as authoritative.

4 L_RAG System Design and Methodology

L_RAG turns the structured corpus and benchmark into an evidence-oriented retrieval and generation pipeline. Offline processing creates normalized records, graph artifacts, and vector or sparse indexes. Online processing resolves query scope, retrieves candidate chunks, optionally expands or filters them with the graph, reranks a bounded pool, and serializes the final context for a citation-grounded generator. The same document, provi-

Table 9: Actual reconciliation counts used by the frozen corpus package.

Validation count	Value
Raw metadata input	153,420
Accepted documents	151,624
Document quarantine	1,796
Raw relationship input	897,890
Accepted relationships	883,256
Relationship quarantine	14,634
Edges with resolved in-corpus target	826,136
Edges with external-stub target	57,120
Unique external stubs	19,763
Text provenance rows	151,624
Text available	127,018
Text missing	5,025
Text empty	19,531
Text too short	50
Provisions	1,386,267
Chunks	1,513,376
Validity events	159,805
Direction-verified validity events	34,379
Direction-unverified validity events	125,426

sion, and chunk identifiers introduced in Section 3 are carried through every stage.

4.1 End-to-End System Architecture

The architecture has five operational phases: (1) query and temporal-scope resolution, (2) dense and optional sparse retrieval, (3) graph filtering or context expansion, (4) rank fusion and candidate reranking, and (5) prompt construction, generation, and cited output. Figure 3 shows the data flow and the point at which each family of ablations changes a component.

The component boundaries are deliberate. The retriever decides which chunks are available, the graph can alter that set under a fixed context budget, the reranker only reorders a candidate pool, and the generator sees a serialized top-10 context. This separation prevents a generation metric from being mistaken for evidence-retrieval quality.

4.2 Offline Data and Index Construction

Offline construction branches from the normalized records into three linked artifacts. The structured-text branch emits provisions and chunks with stable parent pointers. The graph branch emits document, provision, chunk, and relationship nodes plus validity and authority metadata. The retrieval branch embeds either chunk-only or header-plus-chunk text into a FAISS index and builds tokenized BM25 shards for configurations that enable sparse retrieval.

The dense index uses L2-normalized vectors and inner-product search, which is equivalent to cosine ranking for normalized embeddings. For

the completed hybrid runs, BM25 is built as 10 independent shards with shard-local IDF and average-document-length statistics; shard candidates are merged by BM25 score before RRF. The graph and index artifacts keep the same chunk IDs, so a retrieved vector row can be joined back to its document and provision without fuzzy matching. Frozen manifests record the corpus snapshot, model checkpoint, index path, configuration name, and benchmark version used to create each evaluation run.

4.3 Online Query-Processing Workflow

At query time, the system first preserves the raw Vietnamese question and any explicit date or scope constraint. The query encoder produces an E5 vector; the dense branch retrieves a broad seed set, while the optional BM25 branch searches tokenized shards. The branches are merged with RRF when hybrid mode is enabled. In the completed retrieval/graph family, a graph variant keeps the top five seeds, expands them within their parent provisions with `max_hop=1`, and replaces the baseline top-10 context with at most ten expanded chunks. This is a complete context-selection policy: it uses broad validity filtering and does not apply a query-time validity overlay or reranking. The resulting candidates are trimmed to the family-specific budget, reranked only in the reranker family, and serialized as the final context.

Insufficient evidence follows an observable fallback path. If no candidate survives a valid scope filter, if the context contains no usable text, or if the prompt’s evidence threshold is not met, the retrieval/graph prompt asks the model to emit the Vietnamese fallback phrase “*Không có đủ thông tin trong ngữ cảnh được cung cấp.*”; the fixed-context reasoning runs use the visible marker `INSUFFICIENT_CONTEXT`. These strings are saved for evaluation and are not treated as proof that a question was legally unanswerable. Invalid filters and unresolved external references remain visible in the run record instead of being silently repaired.

4.4 Knowledge Graph Construction

The G-LRAG graph uses Document (*D*), Provision (*P*), Chunk (*C*), and ExternalStub nodes. Structural edges encode `DOCUMENT_HAS_PROVISION`, `PROVISION_HAS_CHUNK`, `CHUNK_NEXT`, and `PROVISION_NEXT`. Legal-relation edges encode citations, amendments, replacements, and repeals.

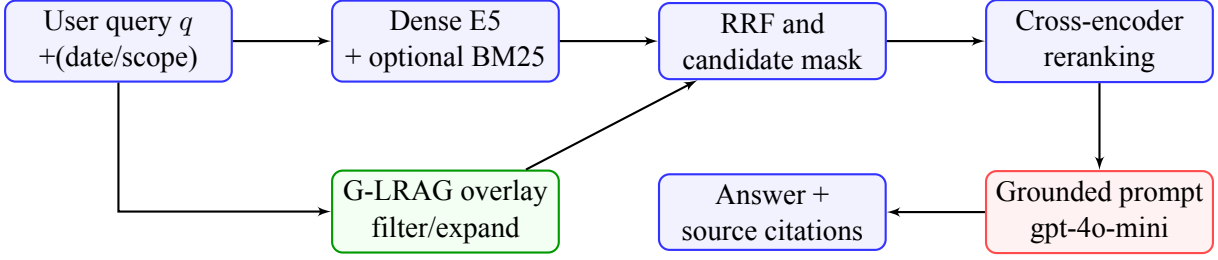


Figure 3: End-to-end operational workflow of L_RAG. The graph and reranker stages are optional branches whose effects are tested under separate frozen ablation controls.

Each edge retains source provenance and is available to query-time traversal only after direction and endpoint checks.

The overlay can fold validity fields relative to an `as_of_date` parameter and can sort candidates by authority rank. A simplified precedence ordering available to the system is

Constitution $\succ$ Law/Code $\succ$ Decree $\succ$ Circular. (4)

Reading-order traversal follows adjacent chunks or provisions to recover nearby context; hop 1 covers the seed provision and additional hops reach subsequent provisions. In the completed graph ablation, traversal is limited to five seeds and at most ten retained chunks, with no query-time validity overlay or reranking. This policy can replace strong seed evidence and is therefore evaluated as a change to the candidate set, not as a free context addition.

4.5 Retrieval and Ranking

The dense encoder is `intfloat/multilingual-e5-large`. For the metadata condition, the indexed string is

$$\text{Text}_{\text{embed}} = \text{DocumentTitle} \parallel \text{ProvisionHeader} \parallel \text{ChunkContent}. \quad (5)$$

the chunk-only condition removes the first two fields. The FAISS index uses `IndexFlatIP`. Sparse retrieval uses `rank_bm25` with `underthesea` tokenization and $k_1 = 1.5$, $b = 0.75$. Dense and sparse lists are fused by

$$\text{RRF}(d) = \sum_{m \in \{\text{dense}, \text{BM25}\}} \frac{1}{60 + r_m(d)}. \quad (6)$$

The dense/hybrid/graph family retrieves 50 seeds and retains at most ten final chunks. The reranker family reconstructs a common RRF pool of 30 candidates (C_{30}), then retains ten (T_{10}).

Table 10: Retrieval and ranking configuration.

Component	Configuration
Dense encoder	<code>intfloat/multilingual-e5-large</code> ; L2-normalized vectors.
Indexed text	Chunk-only or document title + provision header + chunk content.
Dense index	<code>FAISS IndexFlatIP</code> .
Sparse branch	<code>rank_bm25</code> with <code>underthesea</code> ; $k_1 = 1.5$, $b = 0.75$, 10 shards.
Fusion	RRF with $k = 60$ when BM25 is enabled.
Retrieval/graph	50 candidates; 5 graph seeds; <code>max_hop=1</code> ; at most 10 final chunks.
Reranking	Shared C_{30} candidate pool and T_{10} final context; optional cross-encoder.

mMiniLM, Qwen3, BGE, and Jina score the full query–candidate pair with a cross-encoder. Models are loaded, executed, and unloaded sequentially to avoid GPU memory interference. The reranker can improve ordering inside C_{30} , but it cannot recover evidence that never entered the pool.

Table 10 summarizes the retrieval and ranking settings used by the system.

4.6 Answer Generation

The context serializer writes each retained chunk with its chunk ID, document title, provision header, and passage text. All generators use temperature 0.0, but output limits are family-specific: the retrieval/graph runs set no explicit client-side token cap, while the fixed-context reasoning comparison uses a 1,024-token limit. The retrieval/graph prompt asks for Vietnamese answers and uses the fallback phrase “*Không có đủ thông tin trong ngữ cảnh được cung cấp.*” when the evidence is insufficient; the reasoning runs use the visible marker `INSUFFICIENT_CONTEXT`. A citation should identify the governing article or instrument.

The observable output record contains the final answer, citation markers, parser fields, refusal marker, latency, and the ordered context IDs. It does not contain hidden chain-of-thought, reasoning-token accounting, or a human claim-entailment annotation. Consequently, the evalua-

Table 11: Answer-generation configuration.

Setting	Configuration
Generator	<code>gpt-4o-mini</code> ; Base and CoT prompts are compared under fixed context.
Temperature	0.0.
Serialized context	Up to 10 chunks with chunk ID, document title, provision header, and passage text.
Output limit	Retrieval/graph runs: no explicit client-side cap; fixed-context reasoning: 1,024 tokens.
Insufficient evidence	Vietnamese fallback phrase for retrieval/graph; <code>INSUFFICIENT_CONTEXT</code> for reasoning runs.
Recorded output	Answer, citation markers, parser fields, refusal marker, latency, and ordered context IDs.

tion can report output format and lexical overlap but cannot infer legal correctness from a fluent answer or a well-formed citation.

Table 11 records the generator, context, and output controls that determine the observable answer artifacts.

4.7 End-to-End Worked Example

A concrete trace is case `v1rag-5b20bcd0a2` from the completed LLM-BaseReasoning end-to-end run.

Input. “Chúng tôi là đơn vị chuyên môn đang thực hiện phương án phòng chống bệnh dại, nếu kinh phí phát sinh ngoài kế hoạch đã được phê duyệt thì phải làm sao?”

Output. “Nếu có phát sinh về kinh phí so với phương án đã duyệt, UBND tỉnh sẽ xem xét và Sở Tài chính - Vật giá thực hiện theo phê duyệt của UBND tỉnh [1].”

Citation. [1] *V/v tăng cường biện pháp thực hiện phương án phòng chống bệnh dại*, document 116379, item 2, chunk 116379::item::2::chunk::1.

5 Implementation and Experimental Setup

This section states the controls and scoring rules needed to interpret the ablations in Section 6. The benchmark and corpus are shared where possible, but the five reported analyses are grouped into four run families and are not one factorial experiment: candidate pools, context construction, output limits, and latency accounting differ. A result is therefore compared only with its paired control inside the same family.

5.1 Pipeline and Frozen Artifacts

The evaluated pipeline consists of five stages: (1) query embedding with `intfloat/multilingual-e5-large`; (2) dense

FAISS retrieval and, where enabled, tokenized BM25 retrieval; (3) RRF fusion, graph-based context selection, or cross-encoder reranking; (4) serialization of retained chunks into a grounded prompt; and (5) answer generation followed by post-hoc evaluation.

All reported analyses operate on frozen artifacts rather than rerunning models. Each run is associated with a manifest identifying the run configuration, corpus snapshot, benchmark version, model checkpoint or revision, random seed, and code revision. Checksums for the benchmark and major corpus or index artifacts are retained where available so that the evaluated inputs can be distinguished from later repository or dataset updates. Runtime metadata such as execution timestamp, software environment, and hardware configuration is recorded with the run artifacts when available; detailed manifest fields are listed in Appendix F.

The analysis reads saved manifests, per-question predictions, retrieved context identifiers, aggregate metrics, latency records, and error logs. Every primary comparison is aligned by `qa_id`. A row is considered valid when it has status `success`, `completed`, or `ok`, has no saved error, and is not marked `failed`, `error`, or `skipped`; older artifacts without a status field fall back to the presence of a saved prediction. The same identifier links the question category, answer type, reference answer, gold evidence, ordered retrieved context, output diagnostics, and latency record. Reported paired effects are therefore computed only over question IDs satisfying the validity requirements of both compared conditions.

The completed experiments address five research questions:

1. Does document and provision metadata improve exact evidence retrieval or document-level coverage?
2. Does dense-sparse hybrid retrieval improve coverage or rank over dense retrieval alone?
3. Does the current G-LRAG graph policy improve evidence coverage or downstream behavior under a ten-chunk context budget?
4. Which reranker best trades ranking quality against cost when the candidate pool is fixed?
5. Does an explicit chain-of-thought prompt improve observable answer behavior when the retrieved context is fixed?

Table 12: Index, graph, and runtime configuration for completed runs.

Component	Frozen setting
Corpus/index	1,513,376 chunk vectors; graph and index share stable chunk IDs.
Dense retrieval	FAISS IndexFlatIP; 50-result search budget and broad filtering.
Sparse retrieval	10 BM25 shards with shard-local IDF and average-document-length statistics; optional branch.
Fusion	Reciprocal Rank Fusion with $k = 60$ when the sparse branch is enabled.
Graph	Five seed chunks within parent provisions; <code>max_hop=1</code> ; replace baseline context; retain at most 10 chunks.
Reranking	Shared C_{30} pool followed by T_{10} final context; no reranking or one of four cross-encoders.
Generation	<code>gpt-4o-mini</code> ; fixed-context Base/CoT comparison where applicable.

5.2 Indexes, Graph, and Runtime Settings

The dense/hybrid/graph experiment uses a FAISS index with 1,513,376 chunk vectors and a graph over the same chunk identifiers. The hybrid branch searches ten independently built BM25 shards, each with shard-local IDF and average-document-length statistics; shard candidates are merged by BM25 score before RRF. Dense and hybrid retrieval use a 50-result search budget, broad filtering, and a final context of ten chunks; RRF uses $k = 60$ when BM25 is enabled. Graph variants take five seed chunks within their parent provisions, use `max_hop=1`, replace the baseline context, and retain at most ten graph-context chunks. This family uses broad validity filtering without a query-time validity overlay or reranking. The associated generation artifacts use `gpt-4o-mini`.

The reranker family fixes a 30-chunk RRF candidate pool (C_{30}) and a ten-chunk final context (T_{10}). It compares no reranking, RRF-only, mMiniLM, Qwen3, BGE, and Jina. The primary generator study compares GPT-4o-mini Base and CoT with the saved top-10 context fixed by question. The metadata study changes only the indexed string from chunk body to header-plus-chunk text; encoder, FAISS, top- $k = 10$, generator, prompt, seed, corpus, and benchmark remain fixed.

Table 12 summarizes the operational settings used by the completed retrieval, graph, reranking, and generation runs.

5.3 Benchmark and Metrics

The benchmark contains 500 questions: 400 answerable questions with gold chunk identifiers and 100 unanswerable questions. Retrieval metrics are computed only on the 400 answerable questions. Let G_i denote the gold chunk set for question i , and

let $R_{i,k}$ denote the ordered set of retrieved chunks up to rank k . Recall and hit rate are defined as

$$\text{Recall}@k_i = \frac{|G_i \cap R_{i,k}|}{|G_i|}, \quad (7)$$

$$\text{Hit}@k_i = \mathbb{I}[G_i \cap R_{i,k} \neq \emptyset].$$

Let r_i be the rank of the first retrieved gold chunk, with $r_i = \infty$ when no gold chunk is retrieved. Truncated reciprocal rank is

$$\text{MRR}@k_i = \begin{cases} 1/r_i, & r_i \leq k, \\ 0, & r_i > k. \end{cases} \quad (8)$$

For nDCG, retrieved chunks use binary relevance, $\text{rel}_{i,j} = 1$ when the chunk at rank j belongs to G_i and 0 otherwise:

$$\frac{\text{DCG}@k_i}{\text{IDCG}@k_i}, \quad (9)$$

where $\text{IDCG}@k_i$ is the maximum possible DCG obtained by placing up to $\min(|G_i|, k)$ gold chunks at the highest ranks.

Candidate Recall@30 and Candidate Hit@30 measure whether gold evidence is present in the shared 30-candidate pool used by the reranker family. Because this pool is fixed before reranking, these metrics define the maximum evidence coverage available to every reranker condition.

For graph analyses that may return fewer than k chunks, identity-level precision measures the fraction of returned chunks whose mapped document or provision identifier belongs to the corresponding gold identity set. When no chunk is returned, identity precision is defined as zero.

For a normalized prediction y_i and reference answer r_i , Exact Match is

$$\text{EM}_i = \begin{cases} 1, & y_i = r_i, \\ 0, & y_i \neq r_i. \end{cases} \quad (10)$$

The lexical evaluator normalizes text with Unicode NFC, lowercasing, punctuation-to-space replacement, and whitespace collapse. Token F1 uses token overlap O_i and prediction/reference token counts L_i^y and L_i^r when both token lists are non-empty:

$$P_i = \frac{O_i}{L_i^y}, \quad R_i = \frac{O_i}{L_i^r}, \quad F_{1,i} = \frac{2P_iR_i}{P_i + R_i}. \quad (11)$$

When both prediction and reference tokenize to empty strings, Token F1 and ROUGE-L return 1.0;

Table 13: Frozen configurations used by the completed run families. The five reported analyses include separate retrieval and graph views of the joint retrieval/graph family. A dash means that the component was disabled. Candidate/final values differ by family and are not pooled across rows.

Family	Configuration	Dense text	Sparse	Graph	Reranking	Candidate/final
Embedding	Chunk-only	chunk text	—	—	—	10/10
Embedding	Chunk+metadata	header + chunk	—	—	—	10/10
Retrieval/graph	DenseOnly	header + chunk	—	—	—	50/10
Retrieval/graph	HybridOnly	header + chunk	BM25	—	—	50/10
Retrieval/graph	DenseGraph	header + chunk	—	G-LRAG	—	50/10
Retrieval/graph	HybridGraph	header + chunk	BM25	G-LRAG	—	50/10
Reranker	None	shared RRF pool	BM25	—	none	30/10
Reranker	RRF-only	shared RRF pool	BM25	—	RRF	30/10
Reranker	mMiniLM	shared RRF pool	BM25	—	mMiniLM CE	30/10
Reranker	Qwen3	shared RRF pool	BM25	—	Qwen3	30/10
Reranker	BGE	shared RRF pool	BM25	—	BGE-v2-m3	30/10
Reranker	Jina	shared RRF pool	BM25	—	Jina-v2	30/10
LLM reasoning	GPT-4o-mini Base	fixed context	—	—	Base prompt	fixed/10
LLM reasoning	GPT-4o-mini CoT	fixed context	—	—	CoT prompt	fixed/10

when exactly one side is empty, they return 0.0. ROUGE-L otherwise uses the same F-score structure with the token-level longest common subsequence in place of direct token overlap. Exact Match, Token F1, and ROUGE-L are computed only on valid answerable outputs and are treated as lexical-overlap diagnostics rather than measures of legal correctness.

Unanswerable behavior is reported separately from retrieval quality. For the retrieval/graph family, the *fallback-marker rate* is the fraction of the 100 unanswerable outputs containing the expected Vietnamese fallback phrase, “Không có đủ thông tin trong ngữ cảnh được cung cấp.” This is a template-based abstention diagnostic rather than semantic proof that the response correctly recognized insufficient evidence. The 500-question answerability-decision diagnostic used in other run families is reported separately.

Citation presence is the fraction of valid answerable outputs containing at least one parsed citation marker. Structural citation coverage is the fraction of factual sentences identified by the saved output parser that contain at least one citation marker. The post-hoc analysis averages this metric only over valid answerable rows with a non-null saved `structural_citation_coverage` value; rows with no scored factual sentences save this value as null and are excluded from that denominator. These citation metrics measure observable output structure only and do not test whether a cited source entails, fully supports, or legally justifies the associated claim.

Completion rate is the number of valid com-

pleted rows divided by the number of scheduled rows. Latency is reported in seconds using saved mean, p50, and p95 values where available. The hybrid retrieval totals include an amortized BM25 preprocessing cost of 26.990 seconds per query, obtained from 13,495 seconds of batch computation over 500 questions; this value is therefore not a direct online request-latency measurement. Reranker experiments use cached BM25 candidates and likewise do not represent full online sparse-retrieval latency.

Metric preprocessing is evaluator-specific. Retrieval/graph evaluations lowercase text and collapse whitespace while preserving punctuation, whereas the generator-reasoning lexical evaluator applies Unicode NFC normalization and replaces punctuation with spaces before tokenization. Scores are compared only within the evaluator and run family that produced them.

5.4 Pairing, Uncertainty, and Claim Boundaries

Headline effects are paired by `qa_id`. Retrieval deltas use the shared answerable intersection; scheduled-denominator answerability diagnostics retain all 500 cases. Percentile paired-bootstrap confidence intervals use 10,000 resamples (seed 42) for headline metadata, dense/hybrid, reranker, and GPT comparisons, and 2,000 resamples for the smaller reranker slices. An interval crossing zero is reported as inconclusive rather than as evidence of an effect. Exact Match in the primary GPT pair also uses an exact McNemar test.

The primary GPT-4o-mini Base/CoT runs com-

plete 500/500 cases and have identical ordered top-10 chunk-ID sequences for every question. DeepSeek, GLM, and Qwen prompt pairs are descriptive checks: four DeepSeek cases have different context order, and the Qwen manifests differ in code snapshot, timeouts, or retry limits. No prompt effect is pooled across models.

Output limits are family-specific: the retrieval/graph runs set no explicit client-side output-token cap, while the fixed-context generator-reasoning comparison uses a 1,024-token limit. The graph policy is consequently tested as a full context-selection change (seed selection plus expansion), not as an isolated add-on to an otherwise fixed top-10 list.

The artifacts contain no blinded legal-correctness labels, claim-level faithfulness labels, validated semantic judge, preferred answer/clarify/refuse labels, raw chain-of-thought, or reasoning-token accounting. Thus, no result is interpreted as proof of legal correctness, legal safety, citation entailment, or faithful reasoning merely because it improves a lexical, rank, citation-format, refusal-format, or latency metric.

5.4.1 Reproducibility ledger

Each condition is accompanied by a manifest, per-question output records, metric summary, and latency file. Derived analyses are versioned with their source artifacts under `ablation_study/` and `ablation_work/`; the reranker candidate reconstruction and paired analysis are in `ablation_work/reranker_choices/analyze_reranker_metrics.py`, while the generator post-hoc report is in `ablation_study/e2e_LLM_reasoning/results/report.md`.

The report reads these frozen files and does not call a model during analysis.

6 Ablation Studies

We report five completed ablation analyses grouped into four run families: embedding text; dense, hybrid, and graph retrieval as one joint retrieval/graph family; reranking; and generator prompting. Comparisons are made within a family because candidate construction, context budgets, output limits, and latency accounting differ; the results are not treated as one pooled factorial experiment.

6.1 Embedding Text Ablation: Chunk-Only versus Chunk+Metadata

This comparison holds the dense encoder, generator, benchmark, seed, and raw FAISS evaluation path fixed. Only the indexed text changes from the chunk body to a header-plus-chunk representation. On 400 paired answerable questions, exact chunk Recall@10 is effectively unchanged: both variants round to 48.5%, with an unrounded metadata-minus-chunk-only difference of only +0.08 percentage points (reported as +0.1 pp in Table 14). Metadata nevertheless increases document-level success at rank 10 from 65.8% to 70.2%, a paired gain of +4.5 pp (95% CI [+0.5, +8.8]), while MRR@10 decreases by 5.9 pp (95% CI [-9.1, -2.7]).

Because the metadata representation includes both document titles and provision headers, we additionally report provision-level success. At rank 10, provision success is 53.8% for chunk-only retrieval and 53.5% for chunk+metadata, a negligible -0.25 pp difference (95% CI [-5.0, +4.5]). At rank 5, provision success decreases from 44.3% to 40.3%. Thus, the observed metadata gain is concentrated at document discovery rather than provision identification or exact chunk ranking. The retrieval curves and paired outcomes are shown in Figure 4.

6.2 Retrieval Ablation: Dense versus Hybrid Retrieval

The dense/hybrid comparison uses the same header-plus-chunk dense index, 50-seed retrieval budget, 10-chunk final context, benchmark, and generator. Hybrid retrieval adds ten sharded BM25 indexes and reciprocal rank fusion. DenseOnly outperforms HybridOnly on Recall@10 (48.55% versus 42.57%) and nDCG@10 (0.2813 versus 0.2638); the full comparison is in Table 15. The paired Recall@10 difference is -5.98 points with a 95% CI of [-9.71, -2.21]. Hybrid retrieval also raises recorded total median latency from 5.483 seconds to 31.538 seconds. The hybrid total includes 26.990 seconds per query amortized from 13,495 seconds of BM25 batch precomputation over the 500-question batch; this accounting is distinct from online request latency. Paired uncertainty is reported in Table 16.

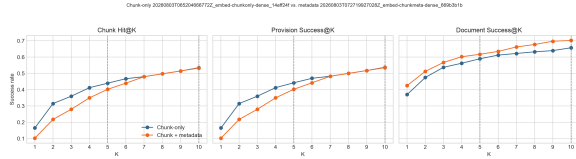


Figure 4: Embedding-text ablation diagnostics. The left panel shows success curves by retrieval depth; the right panel shows paired win, tie, and loss counts. Both plots use the shared answerable-question comparison.

Table 14: Embedding-text ablation on 400 paired answerable questions. Values are percentages. Document and provision success denote retrieval of at least one chunk from a gold document or gold provision, respectively. Deltas are chunk+metadata minus chunk-only. Recall@10 values in the two rows both round to 48.5%; the unrounded difference is +0.08 pp.

Variant	Hit@5	Recall@5	MRR@5	Doc@5	Prov@5	Hit@10	Recall@10	MRR@10	Doc@10	Prov@10
Chunk-only	44.0	39.0	27.4	59.0	44.3	53.5	48.5	28.6	65.8	53.8
Chunk+metadata	40.2	36.8	20.9	61.8	40.3	53.2	48.5	22.7	70.2	53.5
Delta	-3.8	-2.2	-6.5	+2.8	-4.0	-0.2	+0.1	-5.9	+4.5	-0.3

Table 15: Dense versus hybrid retrieval. Retrieval values use 400 answerable questions; latency percentiles use all 500 cases. The fallback column is the fraction of 100 unanswerable outputs containing the family-specific Vietnamese fallback phrase.

Method	Recall@5	Recall@10	MRR@10	nDCG@10	Retrieval p50/p95	Total p50/p95	Fallback
DenseOnly	0.3678	0.4855	0.2267	0.2813	0.797/0.856	5.483/9.946	0.38
HybridOnly	0.3140	0.4257	0.2252	0.2638	27.855/27.924	31.538/33.623	0.38

Table 16: Paired hybrid-minus-dense differences.

Metric	Mean delta	95% CI
Recall@10	-0.0598	[-0.0971, -0.0221]
MRR@10	-0.0015	[-0.0299, +0.0271]
nDCG@10	-0.0175	[-0.0437, +0.0085]

6.3 Graph Ablation: With versus Without Graph Expansion

Graph variants expand the top five seed chunks within their parent provisions with $\text{max_hop}=1$, replace the baseline top-10 context, and retain at most 10 chunks. They use broad validity filtering without a query-time validity overlay or reranking, so this comparison evaluates the complete context-selection policy rather than graph traversal in isolation. Graph expansion currently displaces more evidence than it adds. Relative to DenseOnly, DenseGraph changes Recall@10 by -12.40 points and loses at least one previously retrieved gold chunk for 61 questions while adding a graph-only gold chunk for only one question. The HybridOnly to HybridGraph change is -11.81 points, with 59 questions losing a gold chunk and six gaining one;

the displacement counts and rank deltas are summarized in Table 17.

Graph contexts are denser because they contain fewer chunks on average (5.605 for DenseGraph and 6.305 for HybridGraph), and their identity precision is higher. This is a precision-coverage trade-off, not evidence of an overall retrieval-quality improvement. Of the 65 gold-chunk occurrences lost by DenseGraph, 60 were outside the original top-five seeds; for HybridGraph, 52 of 60 were outside the original top five. The counts support seed-preserving expansion as the next controlled graph test.

6.4 Reranker Ablation

The reranker family compares no reranking, RRF-only, mMiniLM, Qwen3, BGE, and Jina over a reconstructed shared RRF pool of 30 candidates and a final top 10. All six runs cover the same 500 questions without retrieval or generation errors, and the reconstruction matches the recorded RRF-only top-10 prefix for 500/500 questions. Candidate Recall@30 is 0.5995 and Candidate Hit@30 is 0.6450 for every configuration, so the compari-

Table 17: Effect of graph expansion on top-10 gold evidence. Counts are over 400 answerable questions; graph-only gold is introduced by expansion and graph-missed gold was present before expansion but absent afterward. The fallback column is the fraction of 100 unanswerable outputs containing the family-specific Vietnamese fallback phrase.

Base	Δ Recall@10	Δ MRR@10	Δ nDCG@10	Graph-only gold	Graph-missed gold	Avg. chunks	ID precision	Fallback
DenseOnly $\rightarrow$ DenseGraph	-0.1240	-0.0222	-0.0439	1 query (0.25%)	61 queries (15.25%)	5.605	0.0800	0.41
HybridOnly $\rightarrow$ HybridGraph	-0.1181	-0.0240	-0.0411	6 queries (1.50%)	59 queries (14.75%)	6.305	0.0641	0.47

son measures ranking and retention within a fixed evidence ceiling.

Jina is the strongest retrieval configuration: relative to RRF-only, it improves Recall@10 by +9.23 points, MRR@10 by +13.35 points, and nDCG@10 by +11.85 points; all three 95% paired-bootstrap intervals exclude zero. Qwen3 and BGE also have positive intervals, while mMiniLM supplies a smaller positive gain at the lowest reranker cost. BGE has the highest token F1 and ROUGE-L, whereas Jina has the highest observed unanswerable accuracy. These answer metrics are lexical and abstention diagnostics, not legal correctness judgments. Aggregate scores and latency are linked in [Table 18](#), with paired effects in [Table 19](#) and diagnostic curves in [Figure 5](#).

Table 18: Multi-reranker ablation over a shared 30-candidate pool and final top-10 context. Retrieval and overlap values are means over 400 answerable questions; recorded unanswerable accuracy and p50 latency use the complete 500-case runs.

Model	Recall@10	Hit@10	MRR@10	nDCG@10	Token F1	ROUGE-L	Unans. acc.	Rerank p50	Total p50
None	0.4842	0.5375	0.2167	0.2714	0.2801	0.2453	0.856	0.000	2.925
RRF-only	0.4382	0.4800	0.2335	0.2728	0.2750	0.2416	0.874	0.000	2.972
mMiniLM	0.4973	0.5425	0.3047	0.3425	0.2901	0.2568	0.872	0.385	3.343
Qwen3	0.5218	0.5725	0.3418	0.3688	0.2912	0.2583	0.876	9.695	13.007
BGE	0.5188	0.5675	0.3247	0.3593	0.2955	0.2603	0.868	2.872	5.848
Jina	0.5305	0.5850	0.3670	0.3914	0.2913	0.2558	0.884	1.278	4.158

Table 19: Paired effects of each reranker relative to RRF-only. Deltas are in percentage points; intervals are 95% percentile bootstrap CIs from 10,000 question-level resamples over the 400 answerable cases. A negative false refusal delta indicates fewer refusals on answerable questions.

Reranker	Δ Recall@10	Δ MRR@10	Δ nDCG@10	Δ false refusal
mMiniLM	+5.91 [+1.85, +9.96]	+7.12 [+3.55, +10.68]	+6.96 [+3.77, +10.26]	+0.25 [−3.00, +3.50]
Qwen3	+8.37 [+4.57, +12.25]	+10.84 [+7.24, +14.36]	+9.60 [+6.52, +12.73]	−0.25 [−3.25, +2.75]
BGE	+8.07 [+4.29, +11.80]	+9.12 [+5.50, +12.70]	+8.65 [+5.47, +11.84]	+0.75 [−2.25, +3.75]
Jina	+9.23 [+5.51, +12.98]	+13.35 [+9.51, +17.24]	+11.85 [+8.46, +15.27]	−1.25 [−4.00, +1.25]

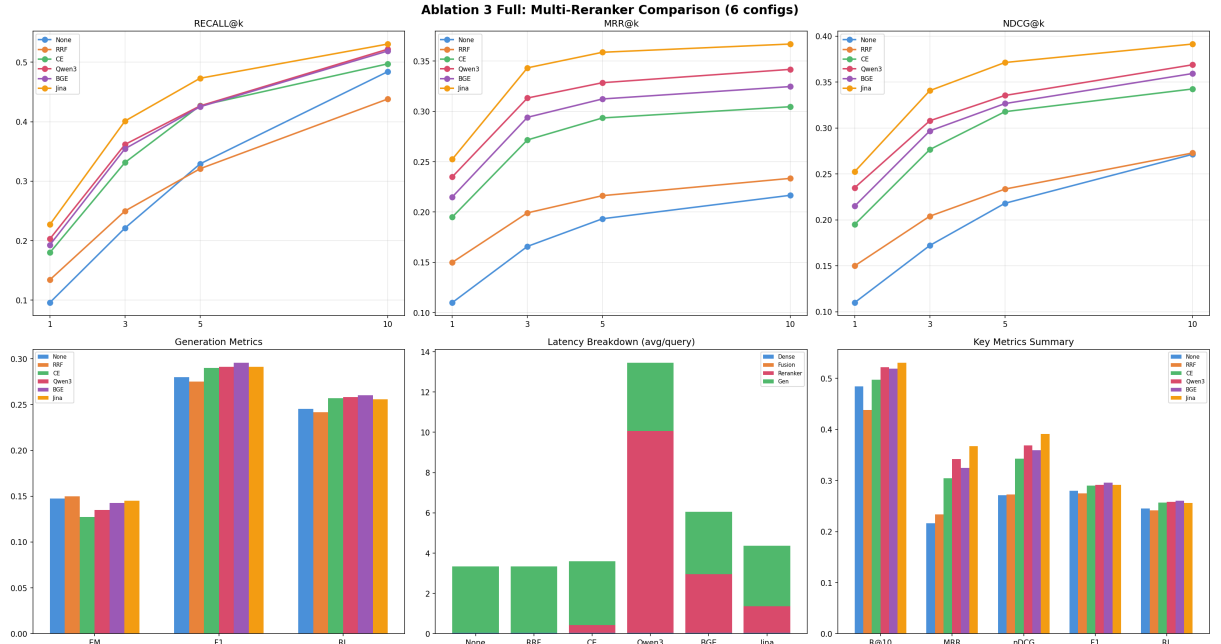


Figure 5: Reranker curves, answer-overlap diagnostics, latency breakdown, and headline metrics from the six-configuration run. The paired confidence intervals in Table 19 are the primary uncertainty analysis.

6.5 Generator Ablation: Base Prompt versus Chain-of-Thought Prompt

6.5.1 Control and pairing

This ablation asks whether an explicit Chain-of-Thought (CoT) prompt changes observable answer behavior when evidence retrieval is held fixed. The pre-specified primary comparison uses GPT-4o-mini with the same benchmark, saved retrieval stack, ordered top-10 context, citation contract, temperature 0, top- $p = 1$, 1,024-token output limit, timeout, retry policy, and seed. Only the prompt strategy and its prompt hash change between Base and CoT. The post-hoc analysis reads the frozen predictions and does not rerun retrieval or generation. The observable prompt and output contracts are reproduced in Appendix E.

The manipulation is deliberately narrow. The Base template asks the generator to answer directly and concisely from the supplied Vietnamese legal passages. The CoT template adds an explicit instruction to inspect and connect the evidence before returning the answer, while retaining the same refusal rule and bracketed citation contract. Evaluation is applied to the visible final answer only. The analysis neither assumes access to provider-side hidden reasoning nor treats any exposed rationale as ground truth; therefore, “CoT” denotes the prompt condition rather than an observed faithful reasoning trace.

Both primary runs complete all 500 scheduled questions without failures or skips. Their ordered top-10 chunk-ID sequences match for all 500 questions, and all 400 answerable cases form valid lexical-metric pairs by `qa_id`; the remaining 100 questions are retained for answerability diagnostics. This establishes a fixed-context comparison: retrieval metrics are controls, not outcomes that the prompt can improve. The complete run inventory is reported in Table 20.

For scoring, the evaluator applies Unicode NFC normalization, lowercasing, punctuation-to-space conversion, and whitespace collapse. Exact Match requires equality after normalization; for Boolean questions it compares the first non-empty line. Token F1 is the harmonic mean of precision and recall over multiset token overlap, while ROUGE-L uses the longest common token subsequence. All deltas below are computed per question as CoT minus Base. Uncertainty is obtained from 10,000 paired bootstrap resamples with seed 42, and the binary Exact Match transitions are additionally checked

with an exact McNemar test.

6.5.2 Primary paired lexical result

On the 400 paired answerable questions, CoT reduces Token F1 from 24.72% to 23.15%, a -1.57 percentage-point change whose 95% paired-bootstrap interval is $[-2.24, -0.92]$. ROUGE-L similarly decreases from 21.60% to 19.90% (-1.71 points, 95% CI $[-2.34, -1.10]$). Both intervals exclude zero. Exact Match changes only from 14.00% to 13.75% (-0.25 points, 95% CI $[-1.50, +1.00]$); its interval includes zero and the exact McNemar test gives $p = 1.00$.

The paired outcomes show that the aggregate decline is not caused by a small number of extreme cases. CoT wins, ties, and loses on Token F1 for 79/164/157 questions and on ROUGE-L for 80/163/157 questions. Thus, losses outnumber wins for both graded overlap metrics, whereas Exact Match is tied on 393 of 400 questions. Table 21 reports the means, paired uncertainty, and per-question win/tie/loss counts.

In relative terms, the Token F1 reduction is 6.35% of the Base mean and the ROUGE-L reduction is 7.92%. These relative values are descriptive—the percentage-point paired effects and their confidence intervals remain the primary quantities—but they show that the change is not negligible compared with the low absolute lexical scores of the task.

6.5.3 Where the paired degradation occurs

The prompt effect is not uniform across benchmark slices. The largest well-populated category decrease occurs for single-hop questions (-2.22 Token-F1 points, $n = 172$), while the citation category also declines (-1.55 points, $n = 106$). The legal-validity interval crosses zero. The cross-document and multi-hop subsets contain only 11 and 20 cases, respectively, so their intervals are reported for transparency but should not be read as stable subgroup estimates. By answer type, both abstractive and extractive answers lose about two Token-F1 points. Boolean answers show a smaller -0.71 -point change whose upper confidence bound is effectively zero.

6.5.4 Evidence, answerability, citation, and latency diagnostics

Conditioning on evidence availability does not reverse the primary result. CoT Token F1 is lower when all annotated support is retrieved (-1.98

Table 20: End-to-end generator-reasoning run inventory. Lexical metrics are means over successful answerable cases; decision accuracy and total p50/p95 latency use all 500 scheduled cases.

Model	Prompt	Run directory	EM (%)	Token F1 (%)	ROUGE-L (%)	Decision acc. (%)	Total p50/p95 (s)
GPT-4o-mini	Base	4o-mini-base	14.00	24.72	21.60	82.60	13.99/17.21
GPT-4o-mini	CoT	4o-mini-CoT	13.75	23.15	19.90	84.00	14.74/18.42
DeepSeek-V3.1-Thinking	Base	deepseekv3.1-thinking-base	22.25	17.84	15.28	92.00	16.15/24.55
DeepSeek-V3.1-Thinking	CoT	deepseekv3.1-thinking-CoT	22.25	16.51	14.12	92.20	18.52/31.37
GLM-5	Base	glm5-base	25.25	16.91	14.90	91.80	14.29/19.09
GLM-5	CoT	glm5-CoT	25.00	16.23	14.27	93.40	15.96/21.23
Qwen3-8B	Base	qwen3-8b-base	19.25	21.98	18.92	95.80	14.15/18.51
Qwen3-8B	CoT	qwen-8b-CoT	17.25	20.05	17.17	95.20	14.58/20.16

Table 21: Primary paired GPT-4o-mini Base-versus-CoT comparison on 400 answerable questions. Deltas are CoT minus Base in percentage points; CIs are percentile bootstrap intervals from 10,000 paired resamples.

Metric	Base (%)	CoT (%)	Δ (pp)	95% paired CI (pp)	CoT W/T/L
Exact Match	14.00	13.75	-0.25	$[-1.50, +1.00]$	3/393/4
Token F1	24.72	23.15	-1.57	$[-2.24, -0.92]$	79/164/157
ROUGE-L	21.60	19.90	-1.71	$[-2.34, -1.10]$	80/163/157

Table 22: GPT-4o-mini Token-F1 slices for the fixed-context Base-versus-CoT comparison. Deltas are CoT minus Base in percentage points; intervals use 10,000 paired bootstrap resamples. Small slices are exploratory.

Slice family	Slice	n	Base	CoT	Δ	95% CI
Category	Citation	106	25.72	24.18	-1.55	$[-2.92, -0.33]$
Category	Cross-document	11	27.67	27.62	-0.05	$[-1.18, +1.05]$
Category	Legal validity	91	14.78	14.38	-0.40	$[-1.89, +1.27]$
Category	Multi-hop	20	30.73	28.33	-2.40	$[-4.22, -0.84]$
Category	Single-hop	172	28.48	26.27	-2.22	$[-3.21, -1.30]$
Answer type	Abstractive	199	32.96	31.02	-1.94	$[-3.09, -0.80]$
Answer type	Boolean	123	8.50	7.78	-0.71	$[-1.46, +0.00]$
Answer type	Extractive	78	29.30	27.31	-1.99	$[-3.39, -0.78]$

points, $n = 138$), when support is partial (-1.86 points, $n = 29$), and when no gold chunk is retrieved (-1.30 points, $n = 233$). ROUGE-L changes in the same direction in all three strata. These are answer-overlap results conditional on a fixed context, not measures of claim faithfulness; the full breakdown is in Table 25.

The saved template-based answerability diagnostic rises from 82.60% to 84.00%. This increase comes entirely from answering seven additional answerable questions: false refusals fall from 85/400 (21.25%) to 78/400 (19.50%), while both prompts correctly refuse 98/100 unanswerable questions. It therefore measures answer/refuse format behavior, not semantic refusal quality. CoT also increases citation presence from 78.25% to 81.00% and mean unique sources from 1.46 to 1.68, but structural citation coverage falls from 89.42% to 87.91%. These citation fields test marker presence and parser coverage rather than source entailment; see Table 24.

The marker counts make the trade-off more concrete. CoT produces 714 parsed citation markers

versus 599 for Base, a 19.2% increase, and neither condition contains an invalid marker. Nevertheless, the lower structural coverage means that adding markers and sources does not ensure that a larger share of scored factual sentences is cited. The data therefore support a formatting claim (more citation activity), not a grounding claim (better evidential support).

A supplementary length diagnostic computed from the frozen final answers also shows that CoT changes response form. On the 400 answerable pairs, mean output length rises from 43.98 to 52.66 whitespace-delimited tokens. The paired mean increase is 8.69 tokens (95% paired-bootstrap CI $[6.41, 11.14]$); CoT is longer on 176 cases, equal on 163, and shorter on 61. These are simple surface tokens rather than provider billing or reasoning tokens. The longer output is consistent with the additional instruction and may contribute to lower lexical overlap, but the saved artifacts do not identify a causal mechanism.

The final column of Table 20 reports saved end-to-end total p50/p95 latency, not generation

time alone. Each percentile uses all 500 rows in the corresponding `latency.json`; the millisecond values were converted to seconds and cross-checked against `run_reliability.csv`. In the controlled GPT-4o-mini pair, generation p50/p95 rises from 2.92/5.74 seconds to 3.68/6.17 seconds, while total p50/p95 rises from 13.99/17.21 seconds to 14.74/18.42 seconds. From the unrounded artifacts, the total median and p95 increases are 0.759 seconds (5.4%) and 1.213 seconds (7.0%). The descriptive p50/p95 increases are 2.37/6.82 seconds for DeepSeek, 1.67/2.13 for GLM-5, and 0.43/1.64 for Qwen3-8B; these secondary pairs are not interpreted as prompt-only effects. The system-wide comparison appears in [Table 26](#).

6.5.5 Descriptive model checks and claim boundary

The other complete model pairs show the same descriptive direction for Token F1: CoT minus Base is -1.33 points for DeepSeek-V3.1-Thinking, -0.68 for GLM-5, and -1.92 for Qwen3-8B. They are not pooled with the primary effect. GPT-4o-mini was the pre-specified primary comparison; four DeepSeek cases have different exact top-10 chunk-ID sequences, and the Qwen manifests use different code snapshots, timeouts, and retry limits.

Under the controlled GPT-4o-mini setting, explicit CoT is therefore not a demonstrated answer-quality improvement: it lowers both graded lexical-overlap metrics and adds latency, while improving only template-based non-refusal and citation-presence diagnostics. The artifacts contain no expert legal labels, claim-level entailment annotations, raw chain-of-thought, reasoning-token counts, or cost records. Accordingly, this ablation supports a conclusion about observable output behavior under a fixed context, not about latent reasoning quality, legal correctness, citation faithfulness, or deployment safety.

6.6 Performance by Question Category and Answer Type

This subgroup analysis is restricted to the 400 answerable questions and compares Jina against RRF-only within the shared reranker candidate pool. Consequently, the slice counts differ from the full benchmark inventory in [Section 3.7](#), which also includes 100 unanswerable questions. The largest observed gains occur for legal-validity and citation questions. The extractive and multi-

hop slices have wider intervals, and the cross-document and hard subsets are too small for stable subgroup estimates. The reported intervals use 2,000 paired-bootstrap resamples; see [Table 23](#).

6.7 Evidence Coverage, Abstention, and Citation Analysis

The GPT-4o-mini pair provides structural diagnostics for evidence availability, answerability decisions, and citations. These fields describe observable output format and saved parser coverage; they do not establish citation entailment, semantic refusal quality, or legal correctness. CoT lowers the false-refusal rate on answerable questions from 21.25% to 19.50%, while unanswerable recognition remains 98.00% for both prompts. Citation presence rises from 78.25% to 81.00% and mean unique sources from 1.46 to 1.68, but structural citation coverage falls from 89.42% to 87.91%; the detailed diagnostics are in [Table 24](#).

Evidence-stratified deltas do not reverse the result: CoT Token F1 is lower when all gold support is retrieved (-1.98 points, $n = 138$), when support is partial (-1.86 points, $n = 29$), and when support is absent (-1.30 points, $n = 233$). The saved records cannot support claim-level faithfulness.

6.8 Latency and Efficiency Comparison

Efficiency differences are substantial. Hybrid retrieval reports a one-time BM25 precomputation cost of about 13,495 seconds, which contributes 26.990 amortized seconds per query to the reported hybrid totals, and a much larger per-query retrieval cost. This batch accounting is not online request latency. Among rerankers, mMiniLM is fastest, Jina is intermediate, BGE is slower, and Qwen3 is much slower. The generator CoT prompt raises generation p50/p95 from 2.92/5.74 seconds to 3.68/6.17 seconds, and total p50/p95 from 13.99/17.21 to 14.74/18.42 seconds. Reranker p95, peak GPU memory, and controlled full-online throughput are unavailable. The complete stage and end-to-end comparison is linked in [Table 26](#).

6.9 Qualitative Ablation Examples and Failure Cases

The saved aggregate artifacts support a precise failure taxonomy even without hand-labelled legal case studies. Metadata has a document-provision trade-off: document-level success increases at top

10 while exact chunk MRR falls, so a retrieved document title is not proof that the governing provision is ranked early. Hybrid fusion can add lexical context while lowering exact gold-chunk coverage; a generation-overlap gain is not evidence that RRF preserved the gold ranked list. Graph expansion has a direct seed-displacement failure: it introduces a new top-10 gold chunk for only 1/400 Dense and 6/400 Hybrid questions, but removes at least one seed-retrieved gold chunk for 61 and 59 questions respectively. Finally, a fluent response with a structural citation marker is not a verified legally correct or entailed answer.

Future case studies should preserve the `qa_id`, gold chunk IDs, ordered candidate IDs before and after the changed component, serialized context, final answer, and citation parser output. A case must label whether its diagnosis is an automatic rank fact or a human legal judgement; this keeps illustrative examples auditable rather than anecdotal.

6.10 Summary of Ablation Findings

The five reported analyses support conditional engineering decisions. Metadata helps document-level retrieval but not exact chunk ranking. The tested hybrid configuration reduces exact-chunk coverage and adds latency. One-hop graph expansion increases context density while displacing more gold evidence than it adds. Cross-encoder reranking is the clearest positive retrieval result: Jina gives the strongest ranking scores, BGE gives the strongest lexical overlap, and mMiniLM gives the lowest recorded reranking cost. For the primary generator pair, CoT lowers lexical overlap under fixed context while improving only template-based decision diagnostics and adding latency. The consolidated recommendations are linked in [Table 27](#).

7 Discussion, Limitations, and Ethical Considerations

7.1 Interpretation of the Main Findings

The ablations yield one clear positive retrieval intervention and several conditional or negative findings. Cross-encoder reranking improves the rank of evidence already admitted to the shared C30 pool; Jina has the strongest retrieval result, while BGE has the strongest lexical answer overlap (the paired evidence is linked in [Table 19](#)). By contrast, the tested hybrid and graph configurations re-

duce exact gold-chunk coverage, as shown by the dense/hybrid comparison in [Table 15](#) and the graph displacement counts in [Table 17](#). These statements concern a finite benchmark and its gold evidence, not legal advice quality in deployment.

7.2 Retrieval Quality, Context Coverage, and Latency Trade-offs

DenseOnly is the preferred retrieval baseline in the dense/hybrid study: it achieves Recall@10 of 0.4855 at 0.797-second retrieval p50. HybridOnly falls to 0.4257 and takes 27.855 seconds when amortized BM25 time is included ([Table 15](#)). That hybrid figure allocates 26.990 seconds per query from a 13,495-second BM25 batch over 500 questions; it should not be read as measured on-line request latency. The graph runs use ten baseline chunks versus five retained seeds before expansion, so their higher identity precision accompanies a smaller context rather than a free addition of evidence. Graph contexts appear more identity-precise because they contain only 5.605–6.305 chunks on average, but their reduced context size also removes gold coverage ([Table 17](#)). The template-based fallback detector also rises from 0.38 to 0.41 for Dense and from 0.38 to 0.47 for Hybrid on the 100 unanswerable questions (baseline rates in [Table 15](#); graph rates in [Table 17](#)). This is an observable refusal-format change, not a validated abstention-quality gain. The reranker result has a different interpretation: it operates after a fixed candidate ceiling, so Jina’s gain is a ranking gain rather than proof that the underlying candidate retriever finds more evidence.

7.3 Implications for Legal RAG Design

For this benchmark, use header-plus-chunk embeddings when correct document identification is more valuable than the rank of an exact chunk; otherwise prefer the chunk-only representation ([Table 14](#)). Do not deploy the current BM25+RRF or one-hop graph setup as the default retrieval path without retuning. Prefer Jina when retrieval quality justifies its 1.278-second reranking p50, or mMiniLM when its lower 0.385-second cost is more important. Retain the Base prompt for lexical answer overlap. In all cases, present citations as sources to inspect and retain human review for legal interpretation.

Table 23: Jina versus RRF-only on selected category and answer-type slices. Deltas are percentage points with 95% paired-bootstrap CIs.

Dimension	Slice	n	Δ Recall@10	Δ MRR@10	Δ nDCG@10
Category	citation	106	+11.89 [+2.82, +21.70]	+16.22 [+7.86, +24.51]	+15.82 [+7.51, +23.82]
Category	legal validity	91	+14.47 [+6.96, +21.43]	+18.90 [+10.28, +27.26]	+15.22 [+9.13, +21.50]
Category	single hop	172	+5.04 [-0.00, +10.37]	+9.63 [+4.34, +14.69]	+8.41 [+3.81, +13.05]
Category	multi hop	20	+7.50 [-0.83, +17.50]	+10.12 [-3.42, +24.33]	+7.95 [-0.77, +17.98]
Answer type	extractive	78	+0.13 [-8.85, +9.74]	+8.53 [-0.90, +17.39]	+7.14 [-1.33, +16.23]
Answer type	abstractive	199	+13.65 [+8.50, +18.80]	+14.74 [+9.61, +20.14]	+13.81 [+9.56, +18.71]
Answer type	boolean	123	+7.86 [+1.63, +14.50]	+14.18 [+7.64, +21.13]	+11.68 [+6.27, +17.95]

Table 24: Structural evidence, abstention, and citation diagnostics for the paired GPT-4o-mini runs.

Diagnostic	Base	CoT	Interpretation
Citation presence	78.25%	81.00%	at least one marker on answerable output
Structural coverage	89.42%	87.91%	saved factual-sentence marker share
Mean unique sources	1.46	1.68	structural source count
False refusal (answerable)	21.25%	19.50%	refusal-marker diagnostic
Unanswerable recognition	98.00%	98.00%	refusal-marker diagnostic

Table 25: Evidence-availability strata for the paired GPT-4o-mini reasoning comparison. Values are lexical means on the 400 paired answerable cases.

Evidence stratum	n	Base F1	CoT F1	Δ F1	Base ROUGE-L	CoT ROUGE-L	Δ ROUGE-L
Support fully retrieved	138	28.24	26.26	-1.98	25.64	23.34	-2.30
Partial support retrieved	29	25.62	23.76	-1.86	21.04	19.45	-1.59
No support retrieved	233	22.53	21.23	-1.30	19.29	17.92	-1.37

Table 26: Recorded latency comparisons. Values are p50/p95 seconds unless noted; reranker p95 was not recorded. Cached BM25 means reranker rows are not full-online sparse-latency measurements.

Configuration	Stage p50/p95	Total p50/p95	Measurement scope
DenseOnly	0.797/0.856	5.483/9.946	measured retrieval + total
DenseGraph	0.798/0.860	4.998/9.750	measured retrieval/graph + total
HybridOnly	27.855/27.924	31.538/33.623	measured + amortized BM25 batch cost
HybridGraph	27.858/27.929	31.180/33.312	measured + amortized BM25 batch cost
mMiniLM	0.385/N/A	3.343/N/A	cached candidate-pool reranking
Jina	1.278/N/A	4.158/N/A	cached candidate-pool reranking
BGE	2.872/N/A	5.848/N/A	cached candidate-pool reranking
Qwen3	9.695/N/A	13.007/N/A	cached candidate-pool reranking
GPT-4o-mini Base	2.92/5.74	13.99/17.21	fixed-context generation
GPT-4o-mini CoT	3.68/6.17	14.74/18.42	fixed-context generation

Table 27: Decision summary for the completed ablation families. Quality claims are limited to the deterministic metrics available in the saved artifacts.

Family	Changed component	Quality effect	Latency effect	Decision
Embedding	chunk text + metadata	document success +4.5 pp; MRR@10 -5.9 pp	not material in headline table	use when document identity is primary
Dense/hybrid	add BM25 + RRF	Recall@10 -5.98 pp	total p50 5.483 s $\rightarrow$ 31.538 s	do not use as default under current settings
Graph	one-hop expansion	Recall@10 -12.40 pp (Dense), -11.81 pp (Hybrid)	graph stage near dense retrieval cost	gate until seed-preserving expansion is tested
Reranker	reorder shared C30 pool	Jina: Recall +9.23 pp, MRR +13.35 pp, nDCG +11.85 pp	mMiniLM fastest; Jina p50 1.278 s	prefer Jina for retrieval quality; mMiniLM for latency
Generator	Base prompt $\rightarrow$ CoT	Token F1 -1.57 pp; ROUGE-L -1.71 pp	total p50 13.99 s $\rightarrow$ 14.74 s	retain Base for this lexical benchmark

7.4 Dataset and Coverage Limitations

The benchmark has 500 questions, of which only 400 are answerable by stored gold chunks. The hard stratum has one answerable item, while multi-hop has 20; those slices are descriptive and cannot establish stable subgroup effects. Gold chunk IDs are an evidence target, not necessarily the full set of legally acceptable authorities. Corpus coverage,

language variation, temporal validity, and external legal references may all limit transfer beyond the data snapshot.

7.5 Knowledge Graph and Validity Limitations

The graph study evaluates a specific policy—five seeds within parent provisions, $\text{max_hop}=1$, broad

validity filtering, no query-time validity overlay or reranking, and ten retained chunks—rather than graph information in general. Its failures point to an engineering hypothesis: expansion substitutes for seeds too early. Of the 65 gold-chunk occurrences lost by DenseGraph, 60 were outside the original top five; the corresponding count for HybridGraph was 52 of 60. These counts and the graph-only recoveries are linked in [Table 17](#). Future graph experiments should preserve high-ranked seeds, separately budget neighbors, verify relationship direction and validity semantics, and measure whether graph-only evidence is legally useful rather than merely novel.

7.6 Retrieval and Generation Limitations

The index, BM25 tokenization, RRF parameter, candidate budgets, rerankers, generator, prompt, and final context size are fixed settings, not optimized universally. The hybrid branch uses shard-local BM25 statistics, and the reported 26.990-second-per-query sparse cost is batch amortization rather than online latency. Reranker timings use cached BM25 candidates and omit full-online sparse retrieval. Lexical overlap may reward reference phrasing rather than legal adequacy. Structural citations measure marker presence or parser coverage, not the relationship between a claim and the cited authority.

7.7 Evaluation Limitations

No blinded legal-correctness labels, claim-level faithfulness labels, or validated semantic judge are available. The report therefore does not claim that any configuration is legally more correct, safer, or more faithful solely because it improves F1, ROUGE-L, citation count, or refusal markers. Full-online p95 reranker latency, peak memory, and throughput are also absent. Non-GPT prompt pairs have provenance or control differences and remain descriptive.

7.8 Ethical Considerations and Safe Use

L_RAG should support research and source discovery, not replace a qualified legal professional. Users should inspect the retrieved authority, its date and validity, and the uncertainty of an answer. Systems should avoid unnecessary personal query data, should not obscure abstentions, and should treat well-cited generated text as a review cue rather than a legal conclusion.

8 Conclusion and Future Work

8.1 Summary of the Implemented System

L_RAG combines a Vietnamese legal chunk corpus, dense FAISS retrieval, optional BM25/RRF fusion, graph expansion, cross-encoder reranking, grounded generation, and saved per-question evaluation records. The ablation report isolates five reported component analyses—embedding text, retrieval mode, graph policy, reranking, and generator prompting—grouped into four run families and evaluated with fixed question sets and paired metrics.

8.2 Main Contributions

The empirical contribution is a traceable set of controlled decisions: the embedded text representation changes document success but not exact chunk coverage; the tested hybrid and graph variants impose measurable coverage losses; reranking within a common pool produces substantial ranking gains; and CoT does not improve lexical answer overlap under fixed context. The report also separates retrieval evidence, answer overlap, refusal formatting, citation structure, and efficiency instead of collapsing them into one score.

8.3 Lessons Learned

The central lesson is that component labels are not outcomes. Metadata, hybrid retrieval, graph expansion, CoT, and cited output are not quality guarantees. Candidate control and paired question-level deltas reveal effects obscured by aggregate scores: metadata can improve document discovery while delaying exact chunks, graph expansion increases density while removing evidence, and rerankers can only improve what has entered C30. The graph loss also reflects a replacement policy that starts from five seeds and retains at most ten expanded chunks; preserving the original seeds is the next targeted test. Frozen manifests and complete row alignment are essential before interpreting a difference as causal.

8.4 Recommended System Configuration

For the evaluated benchmark, the recommended evidence path is dense retrieval with the Jina reranker when its latency is acceptable, a 10-chunk final context, and the Base prompt. Jina is promising within the tested dense-BM25 RRF C_{30} pool; its performance on a dense-only candidate pool remains unmeasured. Use the graph branch only af-

ter seed-preserving expansion is validated, and do not select the current hybrid branch as a default. Its reported latency includes amortized sharded-BM25 batch computation, so online latency must be measured separately. This is a benchmark-bounded configuration, not a deployment approval. A real deployment requires corpus-validity checks, source display, monitored latency, privacy controls, abstention handling, and legal expert review.

8.5 Future Improvements

Priority work is (1) blinded legal-expert annotation for correctness, completeness, and claim-to-citation entailment; (2) seed-preserving graph and neighbor-budget sweeps; (3) BM25/tokenization and RRF-weight tuning under full-online latency measurement; (4) candidate-pool sweeps beyond C30; and (5) temporal-validity and external-reference evaluation. These studies should retain per-case candidates and provenance so failures can be audited.

8.6 Final Conclusion

The ablation evidence supports a practical conclusion: improve the ordering of available legal evidence with reranking before adding unvalidated retrieval complexity. Jina is the strongest observed reranker in the fixed-pool study, while the current hybrid, graph, and CoT variants do not improve their primary quality measures enough to offset their costs or risks. The result is an evidence-based engineering recommendation, bounded by the benchmark and not a claim of legal correctness.

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A Data Record Structure and Core Fields

The corpus and evaluation pipeline use line-delimited JSON records so that individual documents, provisions, chunks, questions, predictions, and runs can be inspected and joined without loading the complete dataset. Table 28 summarizes the core identifiers and representative fields required to reproduce these joins; it should be interpreted as a record inventory rather than a formal JSON Schema specification.

Identifiers are stable across pipeline stages. A chunk resolves to one provision and document, while prediction records join to benchmark questions through `qa_id` and to experimental conditions through `run_id`. Ground-truth evidence is represented by document, provision, and chunk identifier arrays rather than by a text span alone.

For unanswerable questions, the gold evidence arrays are empty by construction. Prediction records retain failed or skipped statuses in the evaluation ledger, whereas answer-overlap metrics use only valid successful answerable rows. These conventions define the joins and evaluation denominators used in Sections 5 and 6.

Appendix B provides compact examples of the corresponding document, provision, chunk, and graph objects.

B Example Documents, Provisions, Chunks, and Edges

The following compact example shows how one source is represented without including a full legal document:

```
document: doc_id=LAW-EXAMPLE-2024
          type=law, effective_date=2024-01-01
provision: provision_id=LAW-EXAMPLE-2024-A15
           article=15, parent=LAW-EXAMPLE-2024
chunk: chunk_id=LAW-EXAMPLE-2024-A15-02
       sequence=2, text="..."
edges: DOCUMENT_HAS_PROVISION -> A15
```

```
PROVISION_HAS_CHUNK -> A15-02
CHUNK_NEXT: A15-01 -> A15-02 -> A15-03
```

The visible passage can be shortened for a prompt, but its parent identifiers remain unchanged. A relation to an unavailable instrument is represented as an external stub rather than a fabricated document:

```
external_stub: stub_id=EXT-8841
               reference="Instrument X, Article 4"
relation: LAW-EXAMPLE-2024 --CITES--> EXT-8841
status: unresolved_target=true
retrievable_text=false
```

These examples make the graph and retrieval joins explicit. They also explain why a high-ranked document can be counted as a document-level success while a specific article or chunk remains absent from the retrieved evidence.

C Graph Relationship Vocabulary

This appendix fixes the interpretation of every edge used by L_RAG. Structural edges are created from the document hierarchy, whereas cross-document edges are normalized from the relationship source. A cross-document record has the following auditable shape:

```
edge_id
src_id, dst_id
rel_canonical, rel_group, rel_raw
direction_normalized, direction_verified
external_target, edge_quality_flags[]
provenance{doc_id, other_doc_id, relationship}
```

The identity key is the full triple (`src_id`, `dst_id`, `rel_canonical`). Consequently, two documents may legitimately be connected by several distinct relationships. The raw Vietnamese label is retained in `rel_raw`; `rel_canonical` is its one-to-one normalized form and is materialized as an upper-snake-case graph type. No canonical label is merged or discarded merely because another label describes its inverse.

C.1 Edge record semantics

Table 29 distinguishes identity, semantics, provenance, and safety controls. This separation is important because a syntactically valid edge is not necessarily eligible for retrieval or temporal reasoning.

C.2 Relationship types and direction

For every normalized edge, the arrow is read literally as “source *REL* destination.” Thus, for example, A –[AMENDS]–> B means that instrument A amends instrument B. Table 30 lists the complete canonical cross-document vocabulary and the structural relations added by the graph builder.

Table 28: Core records used by the corpus and evaluation pipeline.

Record	Required identifiers	Structure	Representative fields
Document	doc_id	JSON object	title, document type, issuing authority, issue/effective dates, validity status, raw-source provenance.
Provision	provision_id, doc_id	JSON object	heading, article/clause/item path, sequence, parent provision.
Chunk	chunk_id, provision_id, doc_id	JSON object	passage text, sequence, embedded-text variant, reading-order pointers.
Question	qa_id	JSON object	question, category, answer type, reference answer, ground-truth document/provision/chunk ID arrays.
Prediction	qa_id, run_id	JSON object	generated answer, ordered context IDs, citation diagnostics, refusal marker, status, and error.
Run manifest	run_id	JSON object	model/checkpoint, corpus snapshot, benchmark/configuration identifiers, seed, code snapshot, and timing artifacts where recorded.

Table 29: Canonical cross-document edge fields and enforcement semantics.

Field	Meaning	Integrity or query-time rule
edge_id	Stable identifier for one normalized relationship row.	Used by audit logs and as the trace target of source_edge_id.
src_id, dst_id	Endpoints in canonical semantic direction.	The builder preserves these endpoints; it does not infer or flip direction.
rel_canonical	Machine-readable one-to-one normalization of the raw label.	Combined with both endpoints to form the full relationship identity.
rel_group	One of basis, citation, related, validity, guidance, supplement, amendment, or suspension.	Selects traversal policy and determines whether direction sign-off is required.
rel_raw	Original Vietnamese relationship label.	Preserved verbatim for audit and mapping review.
direction_normalized	Records that endpoints follow the canonical reading convention.	It is provenance, not permission to reason over an unsigned group.
direction_verified	Human/data-contract sign-off for semantic direction.	Only true edges enter the verified reasoning view.
external_target	Indicates that the target has no complete in-corpus document record.	Requires an external stub; the stub cannot be used as citable evidence.
edge_quality_flags	Non-destructive warnings such as weak relation or missing target metadata.	Retained for reporting and downstream policy; a flag is not silently converted into confidence.
provenance	Original document IDs and relationship surface form.	Allows every normalized edge and derived validity event to be traced upstream.

Table 30: Graph relationship vocabulary, canonical direction, and retrieval role.

Class/group	Canonical relationship(s)	Direction and query-time use
Containment	DOCUMENT_HAS_PROVISION;	Parent document to provision and parent provision to indexed chunk; used to recover citation and hierarchy context.
Reading order	PROVISION_HAS_CHUNK PROVISION_NEXT; CHUNK_NEXT	Earlier sibling to the immediately following sibling; used only for bounded neighboring-context expansion.
basis	BASED_ON	Instrument to the instrument on which it is based; legal-basis traversal is capped at three hops.
citation	CITES	Citing instrument to cited instrument; supplies supplementary evidence rather than a validity decision.
related	RELATED_TO	Source record to a generally related record; symmetric in meaning but retained in its normalized stored direction, and excluded from the default graph-guided filter.
guidance	GUIDES_OR_DETAILS;	Guiding/implementing instrument to governed instrument, or the explicit inverse label; available only after direction sign-off.
supplement	GUIDED_OR_DETAILED_BY SUPPLEMENTS; SUPPLEMENTED_BY	Supplementing instrument to supplemented instrument, or its explicit inverse; represents content addition, not automatic invalidation.
amendment	AMENDS; AMENDED_BY	Amending instrument to amended instrument, or its explicit inverse; represents content change without implying whole-document expiry.
validity	EXPIRES_OR_REPLACES; EXPIRED_OR_REPLACED_BY; PARTIALLY_EXPIRES; PARTIALLY_EXPIRED_BY	Instrument causing whole/partial loss of effect to affected instrument, or the explicit inverse. These edges explain lineage and feed the temporal overlay; they do not directly overwrite currency.
suspension	SUSPENDS; PARTIALLY_SUSPENDS; PARTIALLY_SUSPENDED_BY	SUSPENDED_BY; Suspending instrument to affected instrument, or the explicit inverse; partial relations retain their scope.

C.3 Raw-to-canonical mapping

The dataset contains 17 raw relationship labels. Inverse surface labels remain distinct canonical values, rather than being guessed from endpoint order at query time. The graph type is obtained mechanically by converting the lower-snake-case canoni-

cal value in Table 31 to upper snake case.

This appendix documents the materialized graph artifact and current GraphEdge contract. A separate proposed dataset convention describes folding inverse labels onto a single semantic direction; that convention has not been applied to this

17-type artifact. Adopting it would require an explicit, versioned migration and a new reconciliation of edge identities and counts rather than an undocumented change during retrieval.

C.4 Source, confidence, and exclusion policy

Structural relations inherit the deterministic identities and order fields in `documents.jsonl`, `provisions.jsonl`, and `chunks.jsonl`. Cross-document relations originate in `edges.jsonl`; provenance preserves the original document pair and relationship label so every graph hop can be audited. The v2 reconciliation retains 883,256 normalized edges and quarantines 14,634 rows from 897,890 raw relationships. Quarantine is loss-preserving: rejected rows remain inspectable with their exclusion reasons.

The canonical `GraphEdge` does not invent a scalar confidence score. Confidence is represented by observable controls: the `direction_verified` gate, `edge_quality_flags`, source provenance, and corpus-level quality reports. A normalized but unverified edge may remain in the audit data, but it cannot enter a reasoning path. References to missing in-corpus targets become deterministic `ExternalStub` nodes with `citation_safe=false`; they preserve connectivity but cannot be returned as answer evidence.

The builder consequently exposes two views: all parsed document edges for audit and `verified_document_edges` for reasoning. Duplicate node identifiers are rejected, orphan provisions/chunks and missing stub targets are surfaced as build warnings, and structural-edge counts are reported by type. Query-time traversal must use the verified view; filtering an unsafe edge after generation would be too late because it may already have influenced the answer context.

C.5 Validity and authority conventions

Validity is temporal rather than a permanently stored graph label. Each row in `validity_timeline.jsonl` records `id_str`, `event_type`, `event_date_iso`, `counterparty_id`, `scope`, `rel_canonical`, `source_edge_id`, and `direction_verified`. Event types cover enactment, effectiveness, expiry, replacement, amendment, suspension, and their partial variants. Partial events retain an article-level or free-text scope instead of

collapsing the whole document into one status.

At query time, `currency_status(id_str, as_of_date)` can fold verified events whose effective dates do not exceed an explicitly supplied date. In the current implementation, callers must pass an ISO-formatted `as_of_date` for this guarantee: passing `None` labels the overlay “today” but does not exclude future-dated events. This mismatch is a known implementation limitation, and the reported graph ablation does not claim temporal-applicability evaluation. Every event must trace to a `source_edge_id`. The stored `currency_hint` is a non-authoritative fallback when the timeline is incomplete, and an unresolved case remains unknown. Authority is orthogonal to currency: `authority_index.jsonl` maps document type to a versioned `legal_authority_rank`, where smaller values denote higher authority and 99 denotes an unknown type. Authority, recency, and validity are therefore available as separate, inspectable ranking signals.

The implemented fold sorts date strings with deterministic tie-break fields, discards events whose direction is unverified, and, when an explicit date is provided, ignores later events. ISO-date validation and treatment of undated events must therefore be enforced upstream. The coarse mapping is: enacted/effective to active; expired/replaced to expired; amendment and partial-expiry/replacement events to partial; and whole or partial suspension to suspended. With no eligible event, or with an unrecognized final event, the result is `unknown`. This fold is a retrieval overlay, not a substitute for legal interpretation of a provision’s scope.

For example, consider a hypothetical document with an effective event dated 2020-01-01 and a verified `partially_amended` event dated 2023-07-01 whose scope is “Điều 5.” A query as of 2022-12-31 receives `active`; a query as of 2024-01-01 receives `partial`, while the scope remains available to explain why the document must not be treated as wholly expired. If the amendment edge is unverified, it cannot change the derived status. When multiple authority-index entries match a document type, the implementation selects the lowest rank and then the newest version among equal-rank entries; absence of a match yields rank 99 with an explicit fallback source.

Table 31: Vietnamese source labels and their one-to-one canonical mappings. A slash separates two distinct raw rows and two distinct canonical values.

Raw Vietnamese label(s)	Canonical value(s)	Group
<i>Văn bản căn cứ</i>	based_on	basis
<i>Văn bản dẫn chiếu</i>	cites	citation
<i>Văn bản liên quan khác</i>	related_to	related
<i>Văn bản HD, QĐ chi tiết / Văn bản được HD, QĐ chi tiết</i>	guides_or_details / guided_or_detailed_by	guidance
<i>Văn bản bổ sung / Văn bản được bổ sung</i>	supplements / supplemented_by	supplement
<i>Văn bản sửa đổi / Văn bản được sửa đổi</i>	amends / amended_by	amendment
<i>Văn bản hết hiệu lực / Văn bản quy định hết hiệu lực</i>	expires_or_replaces / expired_or_replaced_by	validity
<i>Văn bản bị hết hiệu lực 1 phần / Văn bản quy định hết hiệu lực 1 phần</i>	partially_expired_by / partially_expires	validity
<i>Văn bản đình chỉ / Văn bản bị đình chỉ</i>	suspends / suspended_by	suspension
<i>Văn bản đình chỉ 1 phần / Văn bản bị đình chỉ 1 phần</i>	partially_suspends / partially_suspended_by	suspension

D Retrieval and Reranker Configurations

Table 32 records the model checkpoints, budgets, and scoring conventions used across the completed runs. It complements the family-level table in Section 5 by documenting implementation details that are useful when reproducing an individual run.

E Prompt and Output Formats

This appendix records the observable contract between retrieval, generation, and evaluation. The primary Base-reasoning comparison fixes the model, retrieved context and order, decoding parameters, and 1,024-token output budget; only prompt_strategy changes. Both conditions use prompt template legal-grounded-answer-v2-citations and citation contract retrieved-context-citations-v1.

E.1 Common prompt envelope

The following is the common, human-readable prompt envelope. Braced fields are filled by the evaluator, and the retrieved passages are treated as evidence, not as further instructions. It consolidates repository-visible instructions; because the frozen manifests retain prompt hashes rather than the complete rendered strings, it is not presented as a byte-for-byte reconstruction. The hashes in Table 33 are the authoritative frozen identities.

Bạn là hệ thống RAG pháp lý Việt Nam.
Chỉ trả lời dựa trên CONTEXT được cung cấp.
Không suy diễn, không bịa thêm ngoài CONTEXT.
Nếu CONTEXT không đủ thông tin để trả lời, hãy trả lời:
"Không có đủ thông tin trong ngữ cảnh được cung cấp."

```
QUESTION:
{question}
ANSWER_TYPE:
{answer_type}
CONTEXT:
{ranked_context}
Trả lời bằng tiếng Việt có dấu:
```

The Base strategy then requests a concise final answer, grounded in the supplied passages, with bracketed citations whenever the context contains a legal basis. The reasoning strategy adds the observable instruction to analyze the supplied evidence before producing the final answer and to attach citations to supported claims. This is a prompt intervention, not evidence that a model’s hidden reasoning is available or faithful. The frozen ablation artifacts contain final answers and output diagnostics, but no raw chain-of-thought, structured justifications, reasoning-token counts, or reasoning cost.

E.2 Prompt variants and frozen identity

The repository-visible strategy-specific instructions are reproduced below in abridged form. The Base condition emphasizes the externally scored answer shape:

Yêu cầu định dạng:
- Nếu answer_type là boolean:
đồng đầu tiên chỉ được là "Có" hoặc "Không";
phần giải thích đặt sau "Giải thích:".
- Nếu answer_type là unanswerable:
chỉ nêu rằng không có đủ thông tin trong ngữ cảnh.
- Với các loại khác:
trả lời ngắn gọn bằng tiếng Việt có dấu;
có thể nêu căn cứ pháp lý nếu CONTEXT có.

The reasoning condition adds an explicit evidence-analysis instruction before the final answer request:

Trước tiên, trình bày quá trình suy luận
(reasoning/thinking) của bạn dựa trên CONTEXT.
Sau đó, đưa ra câu trả lời cuối cùng
bằng tiếng Việt có dấu, kèm căn cứ pháp lý
(citation) nếu có trong CONTEXT.

These excerpts document the observable manipulation; they are not permission to publish or evaluate private model internals. If a provider returns a separate reasoning field, the response parser separates it from the final answer, while the frozen ablation retains and scores only the final-answer surface.

Table 32: System-wide component and model configuration reference.

Component	Model/checkpoint	Key settings	Role and convention
Dense retriever	intfloat/multilingual-e5-large	560M parameters; normalized inner product	Encodes chunk-only or title+header+chunk text in FAISS IndexFlatIP.
Sparse retriever	BM25 (rank_bm25)	$k_1 = 1.5$, $b = 0.75$; underthesea; 10 shards	Sharded lexical search with shard-local IDF/length statistics; candidates merged by score.
Fusion	RRF	$k = 60$; C_{30} where enabled	Combines dense and BM25 ranks before reranking.
Graph overlay	G-LRAG	five seeds; max_hop=1; ten-chunk cap	Structural expansion within parent provisions; retrieval/graph runs use broad validity filtering without query-time overlay or reranking.
mMiniLM	mmarco-mMiniLMv2-L12-H384-v1	117M; FP32	Lightweight cross-encoder baseline.
BGE	BAAI/bge-reranker-v2-m3	567M; multilingual	Cross-encoder with highest observed lexical overlap.
Jina	jinaai/jina-reranker-v2-base-multilingual	278M; position patch	Strongest observed fixed-pool retrieval scores.
Qwen3	Qwen/Qwen3-Reranker-0.6B	0.6B; FP16	CausalLM-based reranker with OOM retry handling.
Generator	gpt-4o-mini	$T = 0.0$; family-specific cap; T_{10}	Retrieval/graph: no explicit client cap and Vietnamese fallback phrase; reasoning: 1,024 tokens and INSUFFICIENT_CONTEXT.

Table 33: Frozen identity of the primary GPT-4o-mini prompt pair. Hashes are SHA-256 prefixes; full values are retained in each run manifest.

Condition	Strategy	Prompt hash	Citation hash	T	top- p	Output cap
Base	base	ccedea197be5	f12b25de1289	0.0	1.0	1,024
CoT	reasoning	a02841e4296c	f12b25de1289	0.0	1.0	1,024

Both manifests also record seed 42, model gpt-4o-mini, template version legal-grounded-answer-v2-citations, and citation-contract version retrieved-context-citations-v1. Identical citation hashes show that citation syntax is a control; different prompt hashes identify the intended intervention. Reproduction should compare these values rather than relying only on directory names.

E.3 Context serialization and citations

Each retrieved chunk is serialized with a stable numeric marker and its legal identity. The implementation uses the citation anchor when present, otherwise the citation label, provision identifier, or chunk identifier in that order:

```
[rank] chunk_id=<chunk_id>;
      provision_id=<provision_id>;
      document_id=<document_id>;
      citation=<citation_anchor>
<chunk_text>
```

The answer cites evidence with [rank] markers, for example [2] or [2] [5]. A marker is valid only when it resolves to an item in the supplied context. The stored citation object preserves document_id, provision_id/section, chunk_id, rank, source path, retrieval score, and the answer offsets of each marker. This makes provision-level citation display reproducible even when several chunks originate from the same instrument.

A schematic two-passage context illustrates the binding between rank markers and stable corpus identities:

```
[1] chunk_id=DOC-A::article::5::chunk::1;
   provision_id=DOC-A::article::5;
   document_id=DOC-A;
   citation=Văn bản A, Điều 5
<text of the first retrieved chunk>

[2] chunk_id=DOC-B::article::2::chunk::1;
   provision_id=DOC-B::article::2;
   document_id=DOC-B;
   citation=Văn bản B, Điều 2
<text of the second retrieved chunk>
```

The numeric marker is local to one prompt and must not be interpreted as a global source identifier. The stable join path is marker $\rightarrow$ context rank $\rightarrow$ chunk_id $\rightarrow$ provision_id $\rightarrow$ document_id. Reordering the same chunks therefore changes marker meaning, which is why the fixed-context ablation verifies the complete ordered top-10 chunk-ID sequence, not merely the set of retrieved documents.

E.4 Answer and refusal formats

An empty context is handled before generation and recorded as a skipped or insufficient-context case. A non-empty but inadequate context must trigger the same refusal rather than a guessed answer. The semantic state may be represented internally as INSUFFICIENT_CONTEXT; it is the Vietnamese sentence above, not that internal token, that forms the visible response contract.

Representative valid surfaces are:

```
boolean:
Có
Giải thích: CONTEXT nêu điều kiện áp dụng tại Điều 5 [1].

extractive/abstractive:
Điều kiện áp dụng gồm ... [1][2].

unanswerable:
Không có đủ thông tin trong ngữ cảnh được cung cấp.
```

For Boolean scoring, only the first non-empty

Table 34: Citation-marker resolution and the diagnostic consequence.

Answer surface	Resolution	Recorded interpretation
... [1]	Marker resolves to context rank 1.	Valid reference; answer offsets and source identity are stored.
... [1] [3]	Both ranks exist in the supplied context.	Two references are stored; unique-source count may still be one if both chunks share a document.
... [11] with top-10 context	No supplied context item has rank 11.	Invalid marker; included in <code>invalid_citation_count</code> and warnings.
Factual sentence without a marker	No local citation reference.	Can reduce structural citation coverage; it is not automatically labeled false.
Refusal sentence with no marker	No evidence asserted.	Parsed through the refusal detector rather than penalized merely for lacking a citation.

Table 35: Observable output convention by benchmark answer type.

Answer type	Visible answer shape	Evaluation convention
boolean	First non-empty line is exactly <i>Có</i> or <i>Không</i> ; supporting text follows <i>Giải thích</i> : and carries citations.	Exact match is computed on the first answer line; faithfulness and citation checks use the complete response.
extractive	Short Vietnamese span or proposition recoverable from the supplied passage, followed by supporting marker(s).	Token F1 and ROUGE-L compare the final answer with the reference; citations are parsed separately.
abstractive	Concise Vietnamese synthesis of the supplied evidence, with markers adjacent to supported factual claims.	Lexical answer metrics and citation diagnostics are both reported; neither alone establishes legal correctness.
unanswerable	Exact visible refusal: “ <i>Không có đủ thông tin trong ngữ cảnh được cung cấp.</i> ”	The evaluator normalizes the refusal into the semantic insufficient-context decision and scores answerability accuracy.

line is used for Exact Match, but citations, refusal detection, and other diagnostics inspect the complete answer. For extractive and abstractive cases, Token F1 and ROUGE-L operate on the normalized final text; they do not validate that a cited source entails the claim. A refusal containing extra unsupported legal advice violates the intent of the refusal contract even if the expected phrase is present.

E.5 Saved prediction record and diagnostic limits

The evaluator stores one auditable record per question. The abbreviated schema below separates the generated text, supplied evidence, parsed citations, and execution state:

```
qa_id, question, category, difficulty, answer_type
reference_answer, predicted_answer, ground_truth
metrics{exact_match, token_f1, rouge_l,
  answerability_decision_accuracy, context_recall@k}
retrieved_context[{rank, document_id, provision_id,
  chunk_id, text, score, citation_anchor}]
citations[{citation_id, document_id, section, chunk_id,
  rank, score, source_path}]
citation_references[{marker, answer_start, answer_end}]
citation_metrics{citation_count, valid_citation_count,
  invalid_citation_count,
  unique_cited_source_count,
  structural_citation_coverage}
status, failed_stage, error, latency_ms
```

The status fields make incomplete runs distinguishable from low-quality answers. An empty retrieved context is intercepted before model invocation; provider or serialization errors populate `failed_stage` and `error`; a successfully parsed response receives `success` even when its lexical score is low. Scheduled-denominator reliability therefore counts failures separately, while

paired lexical analysis requires successful answerable rows in both conditions.

The response parser can detect a provider-supplied reasoning field or remove a `<think>...</think>` block before scoring the final answer; otherwise it records `reasoning_source=not_returned`. For the frozen ablation, only the final-answer surface and observable diagnostics are analyzed. Citation presence, marker validity, unique-source count, and structural coverage measure output structure and parser behavior. They do not prove that a cited provision entails the adjacent claim, that all controlling law was retrieved, or that an abstention was substantively correct.

Reasoning extraction follows a fixed precedence: a dedicated provider field, then a complete `<think>...</think>` block, then `not_returned`. An unterminated think block is not guessed or silently repaired. This parser behavior prevents visible reasoning wrappers from contaminating the scored final answer when a supported response shape is available, but it does not establish that the extracted text is faithful to the model’s internal computation.

F Run Manifests and Additional Results

G Web Demo Interface

The L_RAG prototype includes a web interface for submitting Vietnamese legal questions, inspecting grounded answers, tracing citations to their supporting passages, and handling cases where the

available evidence is insufficient. This appendix illustrates the main user-facing interaction states of the prototype.

Figure 6 shows the initial interface. The user can enter a legal question directly or select one of the suggested examples. The sidebar provides access to the interaction history, Production/Demo mode selection, usage guidance, and configurable retrieval settings. The retrieval scope can also be selected before submitting a question.

Figure 7 illustrates a successful grounded response. The example asks:

“Người lao động được nghỉ phép năm bao nhiêu ngày?”

The interface returns a concise answer stating that an employee who has worked for the same employer for 12 months is entitled to 12 working days of annual leave under normal working conditions. The answer contains an inline citation marker, and the corresponding source is displayed below the response with its document and provision information. This design keeps the generated answer and its supporting legal evidence visible in the same interaction.

Users can inspect a citation in greater detail by opening the source panel. Figure 8 shows the extracted legal passage supplied as evidence to the answer, while Figure 9 presents the corresponding document and provenance metadata.

The interface also exposes explicit insufficient-evidence behavior. Figure 10 shows a question asking whether a specific deadline exists for a facility to complete the purchase of fire-prevention equipment:

“Có thời hạn cụ thể để cơ sở hoàn thành việc mua sắm trang thiết bị phòng cháy, chữa cháy không?”

When the available evidence does not support a reliable answer, the interface displays the state *“Chưa đủ bằng chứng”* rather than presenting an unsupported legal conclusion. No citation is attached to this response because no supporting passage is available for the requested claim.

Together, these interface states expose the main user-facing L_RAG workflow: submitting a legal question, receiving a grounded response, inspecting the cited passage and its provenance, and receiving an explicit abstention when sufficient evidence is unavailable. The interface

therefore makes the distinction between answer generation, evidence inspection, and insufficient-evidence handling visible to the end user.

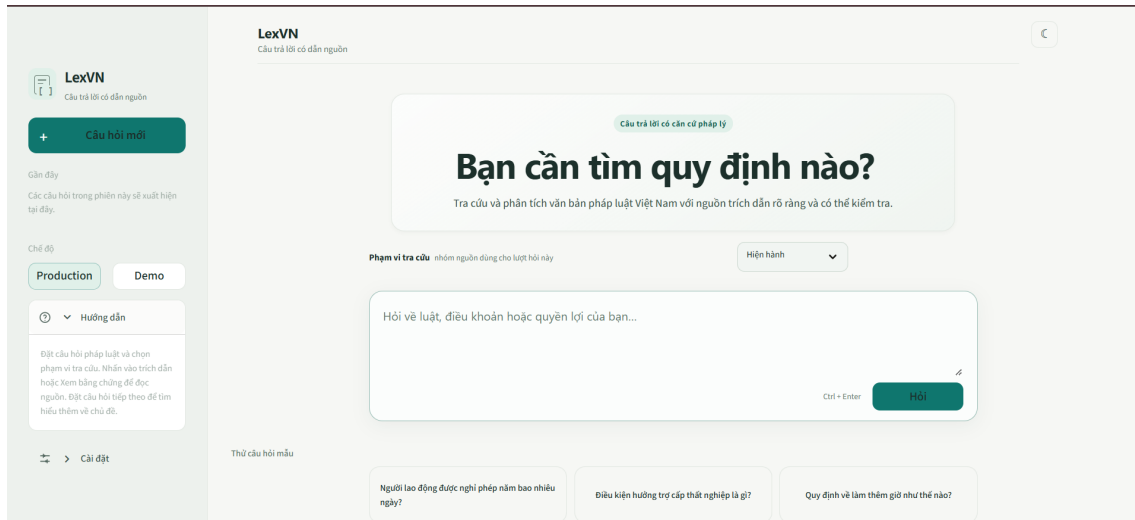


Figure 6: Initial LexVN interface for submitting a Vietnamese legal question. The sidebar exposes session controls, Production/Demo mode, guidance, and retrieval settings, while the main panel provides the query input and suggested questions.

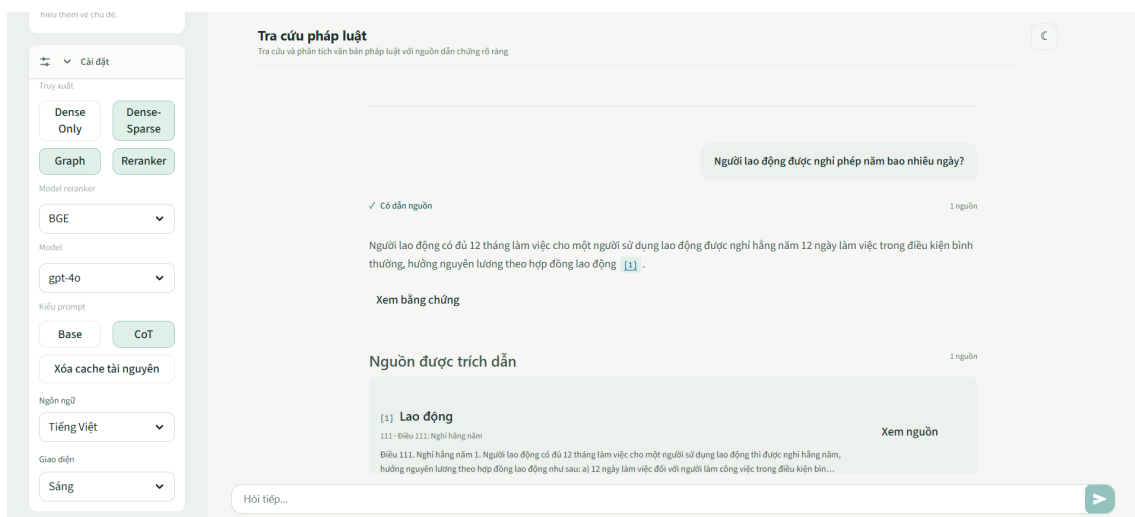


Figure 7: Grounded-answer view for an annual-leave question. The generated answer includes an inline citation, while the supporting source is displayed below for direct inspection.

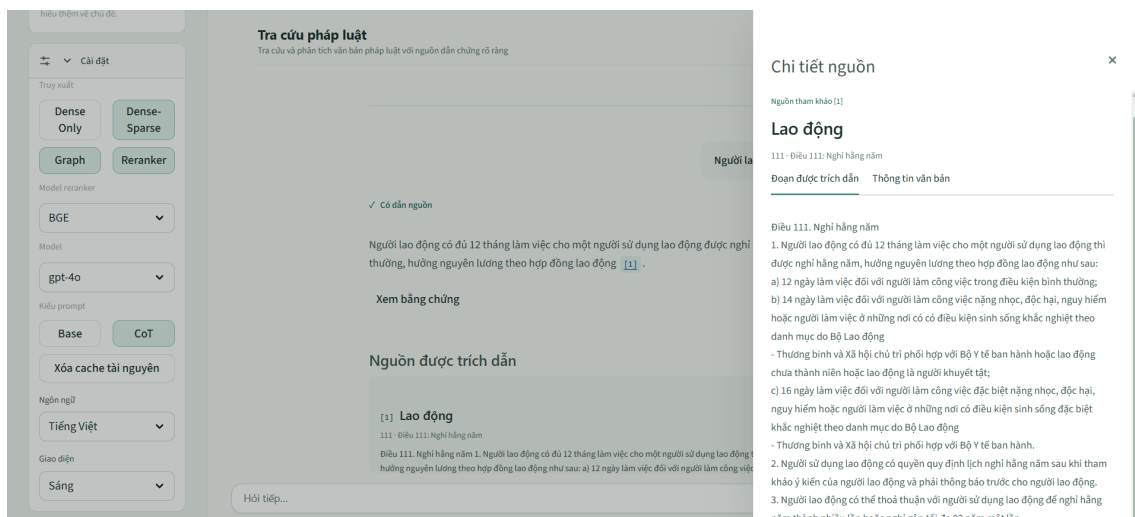


Figure 8: Extracted legal passage associated with the cited source. The panel exposes the text from Article 111 on annual leave that supports the generated answer.

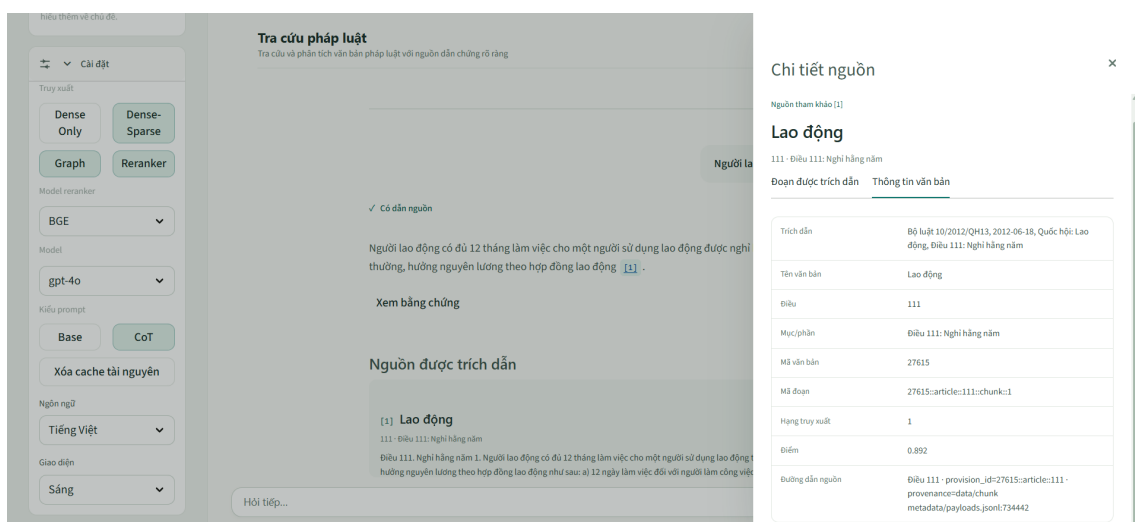


Figure 9: Document and provenance metadata for the cited source, including the document title, provision, chunk identifier, retrieval rank, score, and source path.

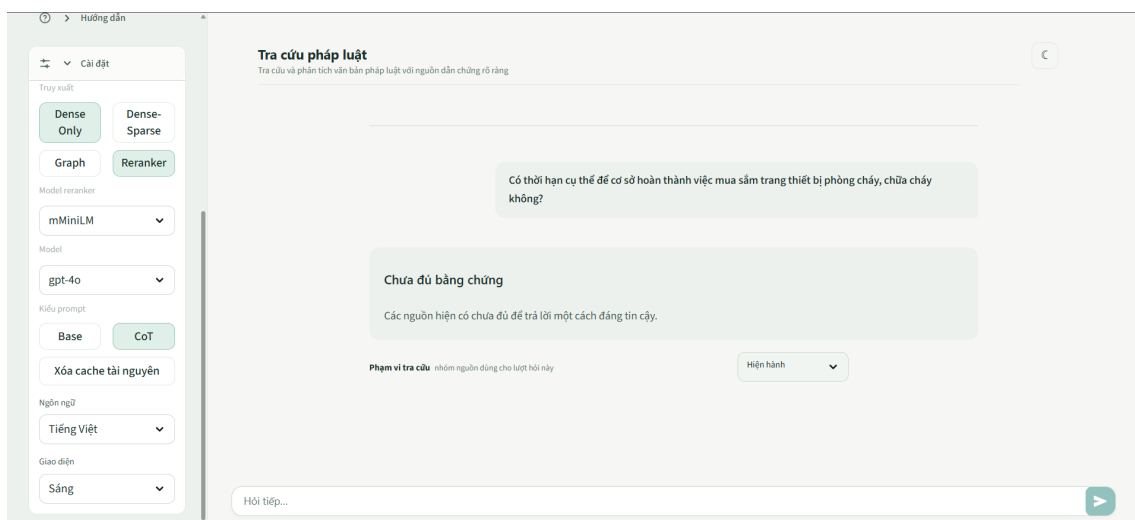


Figure 10: Insufficient-evidence state. When the available sources do not support the requested legal information, the interface explicitly indicates that the evidence is insufficient instead of generating an unsupported answer.